ccna-01-mi-q001 | Basic Questions

Provided HTML: CCNA 200-301 Questions.html | Question 1

What is a function of TFTP in network operations?

A. transfers IOS images from a server to a router for firmware upgrades

B. transfers a configuration files from a server to a router on a congested link

C. transfers files between file systems on a router

D. transfers a backup configuration file from a server to a switch using a username and password

Source answers: A

ccna-01-mi-q002 | Basic Questions

Provided HTML: CCNA 200-301 Questions.html | Question 2

What are two functions of a server on a network? (Choose two)

A. achieves redundancy by exclusively using virtual server clustering

B. handles requests from multiple workstations at the same time

C. runs applications that send and retrieve data for workstations that make requests

D. housed solely in a data center that is dedicated to a single client

E. runs the same operating system in order to communicate with other servers

Source answers: B, C

ccna-01-mi-q003 | Basic Questions

Provided HTML: CCNA 200-301 Questions.html | Question 3

What is the default behavior of a Layer 2 switch when a frame with an unknown destination MAC address is received?

A. The Layer 2 switch drops the received frame

B. The Layer 2 switch floods packets to all ports except the receiving port in the given VLAN

C. The Layer 2 switch forwards the packet and adds the destination MAC address to its MAC address table

D. The Layer 2 switch sends a copy of a packet to CPU for destination MAC address learning

Source answers: B

Source explanation

If the destination MAC address is not in the CAM table (unknown destination MAC address), the switch sends the frame out all other ports that are in the same VLAN as the received frame. This is called flooding. It does not flood the frame out the same port on which the frame was received.

ccna-01-mi-q004 | Basic Questions

Provided HTML: CCNA 200-301 Questions.html | Question 4

What is the same for both copper and fiber interfaces when using SFP modules?

A. They support an inline optical attenuator to enhance signal strength

B. They provide minimal interruption to services by being hot-swappable

C. They accommodate single-mode and multi-mode in a single module

D. They offer reliable bandwidth up to 100 Mbps in half duplex mode

Source answers: B

ccna-01-mi-q005 | Basic Questions

Provided HTML: CCNA 200-301 Questions.html | Question 5

What are two reasons that cause late collisions to increment on an Ethernet interface? (Choose two)

A. when a collision occurs after the 32nd byte of a frame has been transmitted

B. when one side of the connection is configured for half-duplex

C. when the sending device waits 15 seconds before sending the frame again

D. when Carrier Sense Multiple Access/Collision Detection is used

E. when the cable length limits are exceeded

Source answers: B, E

Source explanation

A late collision is defined as any collision that occurs after the first 512 bits (or 64th byte) of the frame have been transmitted. The usual possible causes are full-duplex/half-duplex mismatch, exceeded Ethernet cable length limits, or defective hardware such as incorrect cabling, non-compliant number of hubs in the network, or a bad NIC.

Late collisions should never occur in a properly designed Ethernet network. They usually occur when Ethernet cables are too long or when there are too many repeaters in the network.

Reference: https://www.cisco.com/en/US/docs/internetworking/troubleshooting/guide/tr1904.html

ccna-01-mi-q006 | Basic Questions

Provided HTML: CCNA 200-301 Questions.html | Question 6

In which way does a spine-and-leaf architecture allow for scalability in a network when additional access ports are required?

A. A spine switch and a leaf switch can be added with redundant connections between them

B. A spine switch can be added with at least 40 GB uplinks

C. A leaf switch can be added with a single connection to a core spine switch

D. A leaf switch can be added with connections to every spine switch

Source answers: D

Source explanation

Spine-leaf architecture is typically deployed as two layers: spines (such as an aggregation layer), and leaves (such as an access layer). Spine-leaf topologies provide high-bandwidth, low-latency, nonblocking server-to-server connectivity.

Leaf (access) switches are what provide devices access to the fabric (the network of spine and leaf switches) and are typically deployed at the top of the rack. Generally, devices connect to the leaf switches. Devices can include servers, Layer 4-7 services (firewalls and load balancers), and WAN or Internet routers. Leaf switches do not connect to other leaf switches. In spine-and-leaf architecture, every leaf should connect to every spine in a full mesh.

Spine (aggregation) switches are used to connect to all leaf switches and are typically deployed at the end or middle of the row. Spine switches do not connect to other spine switches.

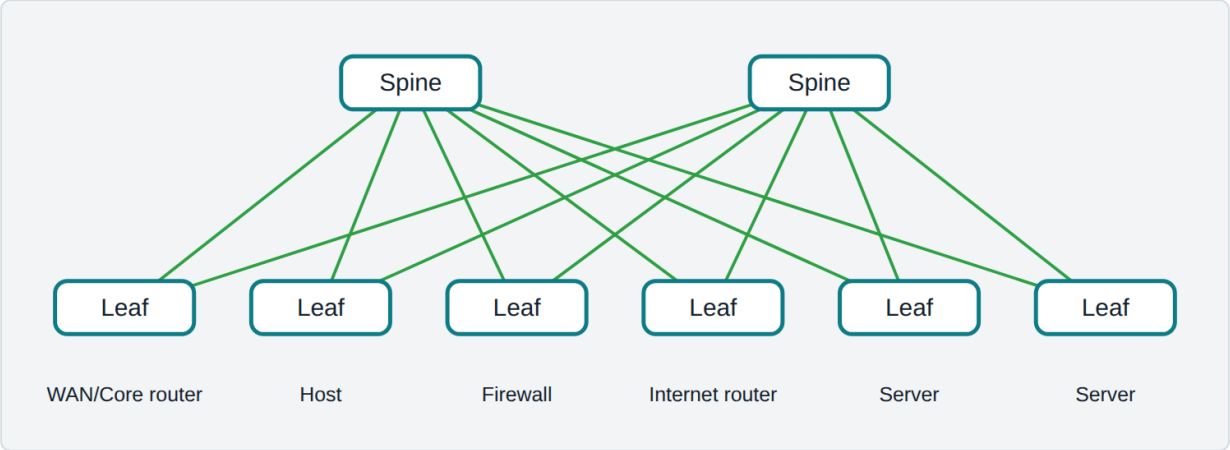
Reference: https://www.cisco.com/c/en/us/products/collateral/switches/nexus-9000-series-switches/guide-c07-733228.html
SpineSpineLeafWAN/Core routerLeafHostLeafFirewallLeafInternet routerLeafServerLeafServer

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Reference: https://www.cisco.com/c/en/us/products/collateral/switches/nexus-9000-series-switches/guide-c07-733228.html



ccna-01-mi-q007 | Basic Questions

Provided HTML: CCNA 200-301 Questions.html | Question 7

What is the function of a server?

A. It creates security zones between trusted and untrusted networks

B. It provides shared applications to end users

C. It routes traffic between Layer 3 devices

D. It transmits packets between hosts in the same broadcast domain

Source answers: B

ccna-01-mi-q008 | Basic Questions

Provided HTML: CCNA 200-301 Questions.html | Question 8

Which two statements about the purpose of the OSI model are accurate? (Choose two)

A. Ensures reliable data delivery through its layered approach

B. Defines the network functions that occur at each layer

C. Facilitates an understanding of how information travels throughout a network

D. Changes in one layer do not impact other layer

Source answers: B, C

ccna-01-mi-q009 | Basic Questions

Provided HTML: CCNA 200-301 Questions.html | Question 9

What is the primary function of a Layer 3 device?

A. to transmit wireless traffic between hosts

B. forward traffic within the same broadcast domain

C. to pass traffic between different networks

D. to analyze traffic and drop unauthorized traffic from the Internet

Source answers: C

ccna-01-mi-q010 | Basic Questions

Provided HTML: CCNA 200-301 Questions.html | Question 10

What is the destination MAC address of a broadcast frame?

A. 00:00:0crfHfrff

B. 00:00:0c:07:ac:01

C. ff:ff:ff:ff:ff:ff

D. 43:2e:08:00:00:0c

E. 00:00:0c:43:2e:08

Source answers: C

ccna-01-mi-q011 | Basic Questions

Provided HTML: CCNA 200-301 Questions.html | Question 11

Which action is taken by a switch port enabled for PoE power classification override?

A. When a powered device begins drawing power from a PoE switch port a syslog message is generated

B. As power usage on a PoE switch port is checked data flow to the connected device is temporarily paused

C. If a switch determines that a device is using less than the minimum configured power it assumes the device has failed and disconnects

D. If a monitored port exceeds the maximum administrative value for power, the port is shutdown and err-disabled

Source answers: D

Source explanation

PoE monitoring and policing compares the power consumption on ports with the administrative maximum value (either a configured maximum value or the port’s default value). If the power consumption on a monitored port exceeds the administrative maximum value, the following actions occur:

+ A syslog message is issued.
+ The monitored port is shut down and error-disabled.
+ The allocated power is freed.

ccna-01-mi-q012 | Basic Questions

Provided HTML: CCNA 200-301 Questions.html | Question 12

A frame that enters a switch fails the Frame Check Sequence. Which two interface counters are incremented? (Choose two)

A. input errors

B. runts

C. giants

D. frame

E. CRC

Source answers: A, E

Source explanation

Whenever the physical transmission has problems, the receiving device might receive a frame whose bits have changed values. These frames do not pass the error detection logic as implemented in the FCS field in the Ethernet trailer. The receiving device discards the frame and counts it as some kind of input error. Cisco switches list this error as a CRC error. Cyclic redundancy check (CRC) is a term related to how the FCS math detects an error.

The “input errors” includes runts, giants, no buffer, CRC, frame, overrun, and ignored counts.

The output below show the interface counters with the “show interface s0/0/0” command:

```
Router#show interface s0/0/0
Serial0/0/0 is up, line protocol is up
Hardware is M4T
Description: Link to R2
Internet address is 10.1.1.1/30
MTU 1500 bytes, BW 1544 Kbit, DLY 20000 usec,
reliability 255/255, txload 1/255, rxload 1/255
--output omitted--
5 minute output rate 0 bits/sec, 0 packets/sec
268 packets input, 24889 bytes, 0 no buffer
Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
251 packets output, 23498 bytes, 0 underruns
0 output errors, 0 collisions, 0 interface resets
0 output buffer failures, 0 output buffers swapped out
0 carrier transitions DCD=up DSR=up DTR=up RTS=up CTS=up
```

ccna-02-mi-q001 | Topology Architecture Questions

Provided HTML: CCNA 200-301 Questions.html | Question 1

Which purpose does a northbound API serve in a controller-based networking architecture?

A. facilitates communication between the controller and the applications

B. reports device errors to a controller

C. generates statistics for network hardware and traffic

D. communicates between the controller and the physical network hardware

Source answers: A

ccna-02-mi-q002 | Topology Architecture Questions

Provided HTML: CCNA 200-301 Questions.html | Question 2

What is a characteristic of spine-and-leaf architecture?

A. It provides greater predictability on STP blocked ports

B. It provides variable latency

C. Each device is separated by the same number of hops

D. Each link between leaf switches allows for higher bandwidth

Source answers: C

Source explanation

Spine-leaf architecture is typically deployed as two layers: spines (such as an aggregation layer), and leaves (such as an access layer). Spine-leaf topologies provide high-bandwidth, low-latency (-> Answer 'It provides variable latency' is not correct), nonblocking server-to-server connectivity.

With a spine-and-leaf architecture, no matter which leaf switch to which a server is connected, its traffic always has to cross the same number of devices to get to another server (unless the other server is located on the same leaf) (-> Answer 'Each device is separated by the same number of hops' is correct) . This approach keeps latency at a predictable level because a payload only has to hop to a spine switch and another leaf switch to reach its destination.

Reference: https://www.cisco.com/c/en/us/products/collateral/switches/nexus-7000-series-switches/white-paper-c11-737022.html

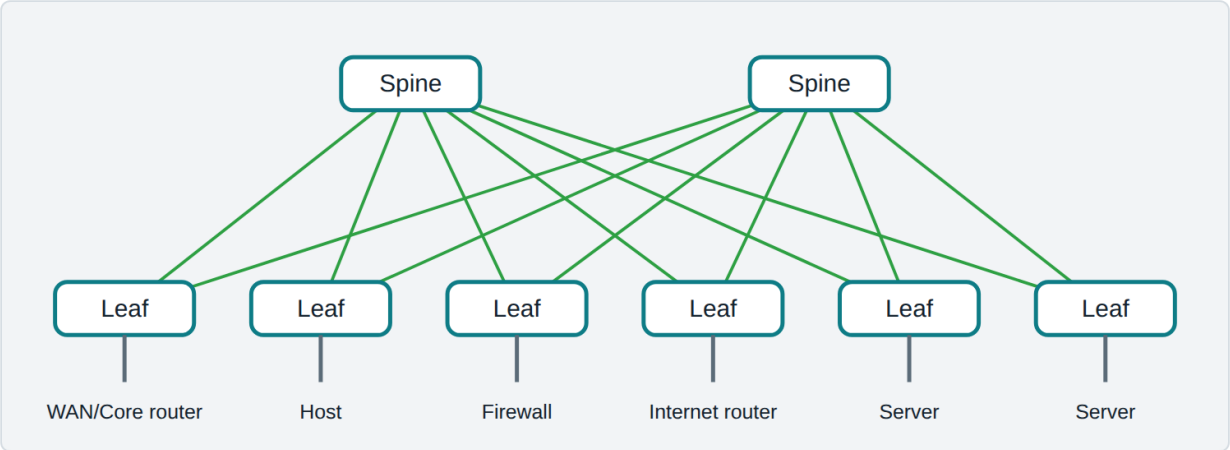
With Leaf-Spine, the network uses Layer 3 routing so STP is no longer required. (-> Answer 'It provides greater predictability on STP blocked ports' is not correct). Spine-leaf architectures rely on protocols such as Equal-Cost Multipath (ECPM) routing to load balance traffic across all available paths while still preventing network loops. This allows all connections to be utilized at the same time while still remaining stable and avoiding loops within the network.
SpineSpineLeafWAN/Core routerLeafHostLeafFirewallLeafInternet routerLeafServerLeafServer

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Reference: https://www.cisco.com/c/en/us/products/collateral/switches/nexus-7000-series-switches/white-paper-c11-737022.html

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ccna-02-mi-q003 | Topology Architecture Questions

Provided HTML: CCNA 200-301 Questions.html | Question 3

Which two functions are performed by the core layer in a three-tier architecture? (Choose two)

A. Inspect packets for malicious activity

B. Provide uninterrupted forwarding service

C. Ensure timely data transfer between layers

D. Police traffic that is sent to the edge of the network

E. Provide direct connectivity for end user devices

Source answers: B, C

Source explanation

Considerations at the core layer include
+ Providing high-speed switching (i.e., fast transport)
+ Providing reliability and fault tolerance
+ Scaling by using faster, and not more, equipment
+ Avoiding CPU-intensive packet manipulation caused by security, inspection, quality of service (QoS) classification, or other processes

Reference: https://www.ciscopress.com/articles/article.asp?p=2202410&seqNum=4

ccna-02-mi-q004 | Topology Architecture Questions

Provided HTML: CCNA 200-301 Questions.html | Question 4

A company needs to interconnect several branch offices across a metropolitan area. The network engineer is seeking a solution that provides high-speed converged traffic, including voice, video, and data on the same network infrastructure. The company also wants easy integration to their existing LAN infrastructure in their office locations. Which technology should be recommended?

A. ISDN

B. Frame Relay

C. VSAT

D. Ethernet WAN

Source answers: D

Source explanation

Ethernet WAN offer the ability of using Ethernet over long-distance links.

ccna-02-mi-q005 | Topology Architecture Questions

Provided HTML: CCNA 200-301 Questions.html | Question 5

What is the name of the layer in the Cisco borderless switched network design that is considered to be the backbone used for high-speed connectivity and fault isolation?

A. access

B. network

C. data link

D. core

E. network access

Source answers: D

Source explanation

The key layers are access, distribution, and core. The core layer is the network backbone that hierarchically connects several layers of the network design, providing for connectivity between end devices, computing and data storage services located within the data center and other areas, and services within the network.

ccna-02-mi-q006 | Topology Architecture Questions

Provided HTML: CCNA 200-301 Questions.html | Question 6

Which two WAN architecture options help a business improve scalability and reliability for the network? (Choose two)

A. dual-homed branches

B. single-homed branches

C. asynchronous routing

D. dynamic routing

E. static routing

Source answers: A, D

Source explanation

Dual-homed branches (branches connects to two ISPs or one ISP with two connections) increases reliability for the network.

Dynamic routing automatically use backup routes in the case of main routes fail which increases reliability. The configuration is remain unchanged or change a little when we expand the network thus increasing the scalability.

Note: A scalable network can expand quickly to support new users and applications without impacting the performance of the service being delivered to existing users

ccna-02-mi-q007 | Topology Architecture Questions

Provided HTML: CCNA 200-301 Questions.html | Question 7

What are two descriptions of three-tier network topologies? (Choose two)

A. The access layer manages routing between devices in different domains

B. The core and distribution layers perform the same functions

C. The core layer maintains wired connections for each host

D. The network core is designed to maintain continuous connectivity when devices fail

E. The distribution layer runs Layer 2 and Layer 3 technologies

Source answers: D, E

ccna-02-mi-q008 | Topology Architecture Questions

Provided HTML: CCNA 200-301 Questions.html | Question 8

What benefit does controller-based networking provide versus traditional networking?

A. allows configuration and monitoring of the network from one centralized port

B. moves from a two-tier to a three-tier network architecture to provide maximum redundancy

C. provides an added layer of security to protect from DDoS attacks

D. combines control and data plane functionality on a single device to minimize latency

Source answers: A

ccna-02-mi-q009 | Topology Architecture Questions

Provided HTML: CCNA 200-301 Questions.html | Question 9

Which function is performed by the collapsed core layer in a two-tier architecture?

A. attaching users to the edge of the network

B. applying security policies

C. marking interesting traffic for data polices

D. enforcing routing policies

Source answers: D

Source explanation

The three-tier hierarchical design maximizes performance, network availability, and the ability to scale the network design.
However, many small enterprise networks do not grow significantly larger over time. Therefore, a two-tier hierarchical design where the core and distribution layers are collapsed into one layer is often more practical. A “collapsed core” is when the distribution layer and core layer functions are implemented by a single device. The primary motivation for the collapsed core design is reducing network cost, while maintaining most of the benefits of the three-tier hierarchical model.

Reference: https://www.ciscopress.com/articles/article.asp?p=2202410&seqNum=4

A collapsed core network is shown below. The collapsed core network may be deployed with redundant core/distribution router, or consolidated core/distribution router.

Deploying a collapsed core network results in the distribution layer and core layer functions being implemented in a single device. The collapsed core/distribution device must provide the following:
+ High speed physical and logical paths connecting to the network
+ Layer-2 aggregation and demarcation point
+ Define routing and network access policies -> Answer 'enforcing routing policies' is correct
+ Intelligent network services—QoS, Network virtualization, etc.

Reference:
https://www.cisco.com/c/en/us/td/docs/solutions/Enterprise/Small_Enterprise_Design_Profile/SEDP/chap2.html

ccna-02-mi-q010 | Topology Architecture Questions

Provided HTML: CCNA 200-301 Questions.html | Question 10

Which WAN access technology is preferred for a small office / home office architecture?

A. Integrated Services Digital Network switching

B. frame-relay packet switching

C. dedicated point-to-point leased line

D. broadband cable access

Source answers: D

ccna-04-mi-q001 | CDP & LLDP Questions

Provided HTML: CCNA 200-301 Questions.html | Question 1

Refer to the exhibit. Which command provides this output?

```
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
 S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone,
 D - Remote, C - CVTA, M - Two-port Mac Relay
Device ID  Local Intrfce  Holdtme  Capability  Platform  Port ID
10.1.1.2   Gig 37/3       176      R I         CPT 600   Gig 36/41
10.1.1.2   Gig 37/1       174      R I         CPT 600   Gig 36/43
10.1.1.2   Gig 36/41      134      R I         CPT 600   Gig 37/3
10.1.1.2   Gig 36/43      134      R I         CPT 600   Gig 37/1
10.1.1.2   Ten 3/2        132      R I         CPT 600   Ten 4/2
10.1.1.2   Ten 4/2        174      R I         CPT 600   Ten 3/2
```

A. show ip route

B. show ip interface

C. show interface

D. show cdp neighbor

Source answers: D

ccna-04-mi-q002 | CDP & LLDP Questions

Provided HTML: CCNA 200-301 Questions.html | Question 2

An engineer needs to configure LLDP to send the port description time length value (TLV). What command sequence must be implemented?

A. switch#lldp port-description

B. switch(config)#lldp port-description

C. switch(config)#interface gi1/0/1
switch(config-if)#lldp port-description

D. switch#interface gi1/0/1
switch(config-if)#lldp port-description

Source answers: B

Source explanation

Use the switch(config)#lldp port-description option to specify the port description TLV messages.

ccna-04-mi-q003 | CDP & LLDP Questions

Provided HTML: CCNA 200-301 Questions.html | Question 3

A network engineer must create a diagram of a multivendor network. Which command must be configured on the Cisco devices so that the topology of the network can be mapped?

A. Device(Config)#lldp run

B. Device(Config)#cdp run

C. Device(Config-if)#cdp enable

D. Device(Config)#flow-sampler-map topology

Source answers: A

ccna-04-mi-q004 | CDP & LLDP Questions

Provided HTML: CCNA 200-301 Questions.html | Question 4

In a CDP environment, what happens when the CDP interface on an adjacent device is configured without an IP address?

A. CDP operates normally, but it cannot provide any information for that neighbor.

B. CDP operates normally, but it cannot provide IP address information for that neighbor.

C. CDP uses the IP address of another interface for that neighbor.

D. CDP becomes inoperable on that neighbor.

Source answers: C

Source explanation

If a neighbor has no IP address on an interface enabled with Cisco Discovery Protocol, the IP address of another interface will be updated as IP address for the non-IP address interface.

Reference: https://www.cisco.com/c/en/us/td/docs/ios-xml/ios/cdp/configuration/15-mt/cdp-15-mt-book/nm-cdp-discover.html

ccna-04-mi-q005 | CDP & LLDP Questions

Provided HTML: CCNA 200-301 Questions.html | Question 5

How can the Cisco Discovery Protocol be used?

A. to allow a switch to discover the devices that are connected to its ports

B. to determine the hardware platform of the device

C. to determine the IP addresses of connected Cisco devices

D. all of the above

Source answers: D

ccna-04-mi-q006 | CDP & LLDP Questions

Provided HTML: CCNA 200-301 Questions.html | Question 6

Which command is used to specify the delay time in seconds for LLDP to initialize on any interface?

A. lldp timer

B. lldp holdtime

C. lldp reinit

D. lldp tlv-select

Source answers: C

Source explanation

+ lldp holdtime seconds: Specify the amount of time a receiving device should hold the information from your device before discarding it
+ lldp reinit delay: Specify the delay time in seconds for LLDP to initialize on an interface
+ lldp timer rate: Set the sending frequency of LLDP updates in seconds

Reference: https://www.cisco.com/c/en/us/td/docs/switches/lan/catalyst3560/software/release/12-2_55_se/configuration/guide/3560_scg/swlldp.html

ccna-04-mi-q007 | CDP & LLDP Questions

Provided HTML: CCNA 200-301 Questions.html | Question 7

Which command should you enter to configure an LLDP delay time of 5 seconds?

A. lldp reinit 5

B. lldp reinit 5000

C. lldp holdtime 5

D. lldp timer 5000

Source answers: A

Source explanation

+ lldp holdtime seconds: Specify the amount of time a receiving device should hold the information from your device before discarding it
+ lldp reinit delay: Specify the delay time in seconds for LLDP to initialize on an interface
+ lldp timer rate: Set the sending frequency of LLDP updates in seconds

Reference: https://www.cisco.com/c/en/us/td/docs/switches/lan/catalyst3560/software/release/12-2_55_se/configuration/guide/3560_scg/swlldp.html

ccna-04-mi-q008 | CDP & LLDP Questions

Provided HTML: CCNA 200-301 Questions.html | Question 8

Which command is used to enable LLDP globally on a Cisco IOS ISR?

A. lldp run

B. lldp enable

C. lldp transmit

D. cdp run

E. cdp enable

Source answers: A

Source explanation

Link Layer Discovery Protocol (LLDP) is a industry standard protocol that allows devices to advertise, and discover connected devices, and there capabilities (same as CDP of Cisco). To enable it on Cisco devices, we have to use this command under global configuration mode:

Switch(config)# lldp run

ccna-05-mi-q001 | Switch Questions

Provided HTML: CCNA 200-301 Questions.html | Question 1

What are two functions of a Layer 2 switch? (Choose two)

A. selects the best route between networks on a WAN

B. moves packets between different VLANs

C. acts as a central point for association and authentication servers

D. moves packets within a VLAN

E. makes forwarding decisions based on the MAC address of a packet

Source answers: D, E

ccna-05-mi-q002 | Switch Questions

Provided HTML: CCNA 200-301 Questions.html | Question 2

The SW1 interface g0/1 is in the down/down state. Which two configurations are valid reasons for the interface conditions? (Choose two)

A. The interface is error-disabled

B. The interface is shut down

C. There is a protocol mismatch

D. There is a duplex mismatch

E. There is a speed mismatch

Source answers: A, E

Source explanation

The down/down state indicates a Layer 1 problem -> “Protocol mismatch” answer is a Layer 2 problem so it is not correct.

Duplex should be matched on both side but it is not a must -> Therefore “duplex mismatch” is not a correct answer.

The most common Layer 1 problems are faulty or incorrect cabling or hardware failure but none of them is in the answer choices.

If the port is error-disabled, we will see it in down/down state so this answer is correct.

SW#sh int f1/0/1
FastEthernet1/0/1 is down, line protocol is down (err-disabled)

There are only two answers left, which are “speed mismatch” and “interface is shut down”. According to the table below (from the old and retired ICND1 100-101 Official Cert Guide), speed mismatch is also a cause of down/down state.

Table 10-4 LAN Switch Interface Status Codes
| Line Status | Protocol Status | Interface Status | Typical Root Cause

| Administratively Down | Down | disabled | The interface is configured with the shutdown command.

| Down | Down | notconnect | No cable; bad cable; wrong cable pinouts; the speeds are mismatched on the two connected devices; the device on the other end of the cable is (a) powered off, (b) shutdown, or (c) error disabled.

| Up | Down | notconnect | An interface up/down state is not expected on LAN switch physical interfaces.

| Down | Down (err-disabled) | err-disabled | Port security has disabled the interface.

| Up | Up | connected | The interface is working.

Note:
+ Speed must match between two ends. We tested speed mismatch on real device but we only received “up/down” state.
+ If “interface is shut down” then we will see the “administratively down/down”, not “down/down” state. But the “down/down” state would be seen on the far end interface.

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Down	Down (err-disabled)	err-disabled	Port security has disabled the interface.
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ccna-05-mi-q003 | Switch Questions

Provided HTML: CCNA 200-301 Questions.html | Question 3

How does the dynamically-learned MAC address feature function?

A. It requires a minimum number of secure MAC addresses to be filled dynamically

B. The ports are restricted and learn up to a maximum of 10 dynamically-learned addresses

C. Switches dynamically learn MAC addresses of each connecting CAM table

D. The CAM table is empty until ingress traffic arrives at each port

Source answers: D

Source explanation

The MAC addresses in the CAM table are the source MAC addresses only. Therefore it only learns MAC address from ingress traffic.

ccna-05-mi-q004 | Switch Questions

Provided HTML: CCNA 200-301 Questions.html | Question 4

Which three statements about MAC addresses are correct? (Choose three)

A. The MAC address of a device must be configured in the Cisco IOS CLI by a user with administrative privileges

B. A MAC address contains two main components, the first of which identifies the network on which the host resides and the second of which uniquely identifies the host on the network

C. A MAC address contains two main components, the first of which identifies the manufacturer of the hardware and the second of which uniquely identifies the hardware

D. An example of a MAC address is 0A:26:B8:D6:65:90

E. To communicate with other devices on a network, a network device must have a unique MAC address

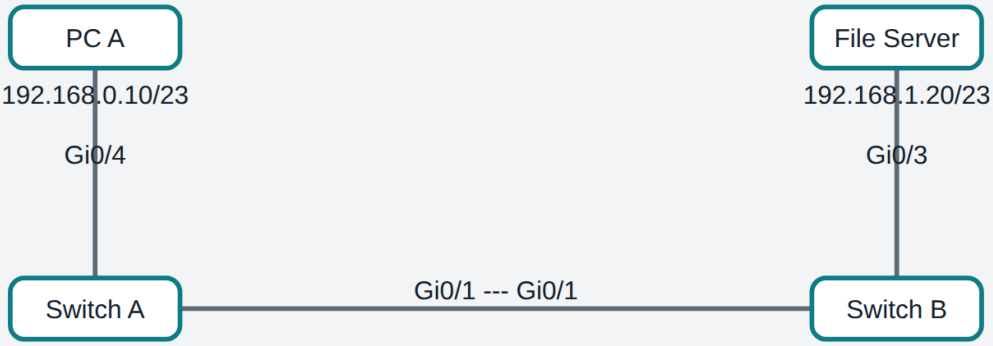
F. The MAC address is also referred to as the IP address

Source answers: C, D, E

ccna-07-mi-q001 | VLAN & Trunking Questions 2

Provided HTML: CCNA 200-301 Questions.html | Question 1

Refer to the exhibit. A network administrator assumes a task to complete the connectivity between PC A and the File Server. Switch A and Switch B have been partially configured with VLANs 10, 11, 12 and 13. What is the next step in the configuration?




```
Switch A
Vlan 10,11,12,13

interface GigabitEthernet0/1
 switchport mode trunk
 switchport trunk allowed vlan 10-12
!
interface GigabitEthernet0/4
 switchport access vlan 13
 switchport mode access

Switch B
Vlan 10,11,12,13

interface GigabitEthernet0/1
 switchport mode trunk
!
interface GigabitEthernet0/3
 switchport access vlan 13
 switchport mode access
```

A. Add a router on a stick between Switch A and Switch B allowing for inter-VLAN routing

B. Add VLAN 13 to the trunk links on Switch A and Switch B for VLAN propagation

C. Add PC A to VLAN 10 and the File Server to VLAN 11 for VLAN segmentation

D. Add PC A to the same subnet as the File Server allowing for intra-VLAN communication

Source answers: B

Source explanation

In fact we only need to add (allow) VLAN 13 to the trunk link of Switch A. Switch B allows all VLANs by default so we don’t need to do any further configuration.

ccna-07-mi-q002 | VLAN & Trunking Questions 2

Provided HTML: CCNA 200-301 Questions.html | Question 2

An engineer must establish a trunk link between two switches. The neighboring switch is set to trunk or desirable mode. What action should be taken?

A. configure switchport trunk dynamic desirable

B. configure switchport mode dynamic desirable

C. configure switchport mode dynamic auto

D. configure switchport nonegotiate

Source answers: B

Source explanation

If the neighboring switch is set to trunk then we need to set our switch to desirable mode so that it can start the negotiation. In fact answer "configure switchport mode dynamic auto" is also correct according to the table below:

| | Dynamic Auto | Dynamic Desirable | Trunk | Access

| Dynamic Auto | Access | Trunk | Trunk | Access

| Dynamic Desirable | Trunk | Trunk | Trunk | Access

| Trunk | Trunk | Trunk | Trunk | Limited Connectivity

| Access | Access | Access | Limited Connectivity | Access

But “desirable” mode is a safer answer so we should choose it.

> If the neighboring switch is set to trunk then we need to set our switch to desirable mode so that it can start the negotiation. In fact answer "configure switchport mode dynamic auto" is also correct according to the table below:

	Dynamic Auto	Dynamic Desirable	Trunk	Access
Dynamic Auto	Access	Trunk	Trunk	Access
Dynamic Desirable	Trunk	Trunk	Trunk	Access
Trunk	Trunk	Trunk	Trunk	Limited Connectivity
Access	Access	Access	Limited Connectivity	Access

> But “desirable” mode is a safer answer so we should choose it.

ccna-07-mi-q003 | VLAN & Trunking Questions 2

Provided HTML: CCNA 200-301 Questions.html | Question 3

An engineer requires a switch interface to actively attempt to establish a trunk link with a neighbor switch. What command must be configured?

A. switchport mode trunk

B. switchport mode dynamic auto

C. switchport nonegotiate

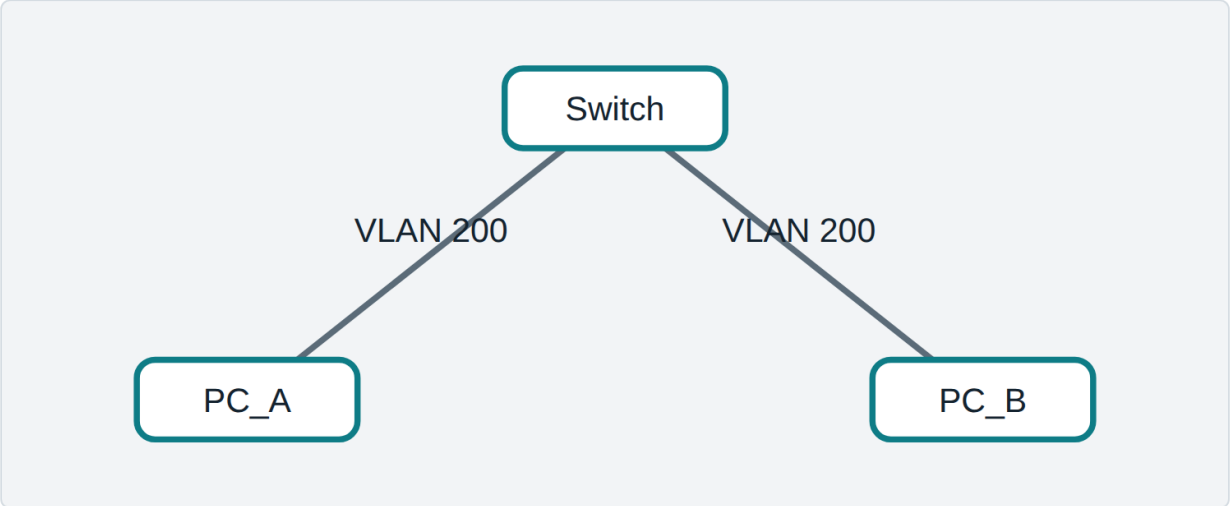
D. switchport mode dynamic desirable

Source answers: D

ccna-07-mi-q004 | VLAN & Trunking Questions 2

Provided HTML: CCNA 200-301 Questions.html | Question 4

Refer to the exhibit. Which outcome is expected when PC_A sends data to PC_B?



A. The source MAC address is changed

B. The destination MAC address is replaced with ffff.ffff.ffff

C. The source and destination MAC addresses remain the same

D. The switch rewrites the source and destination MAC addresses with its own

Source answers: C

Source explanation

Layer 2 switches do not change MAC address so PC_A will put PC_B MAC address into its destination address before sending it to the switch. PC_B will receive the frames with the source and destination MAC address intact.

Note: Only with routers, including layer-3 switches where the packets need to cross to other VLANs, will the frames be stripped and rewritten for the new network or VLAN.

ccna-07-mi-q005 | VLAN & Trunking Questions 2

Provided HTML: CCNA 200-301 Questions.html | Question 5

An engineer must configure interswitch VLAN communication between a Cisco switch and a third-party switch. Which action should be taken?

A. configure IEEE 802.1q

B. configure IEEE 802.1p

C. configure DSCP

D. configure ISL

Source answers: A

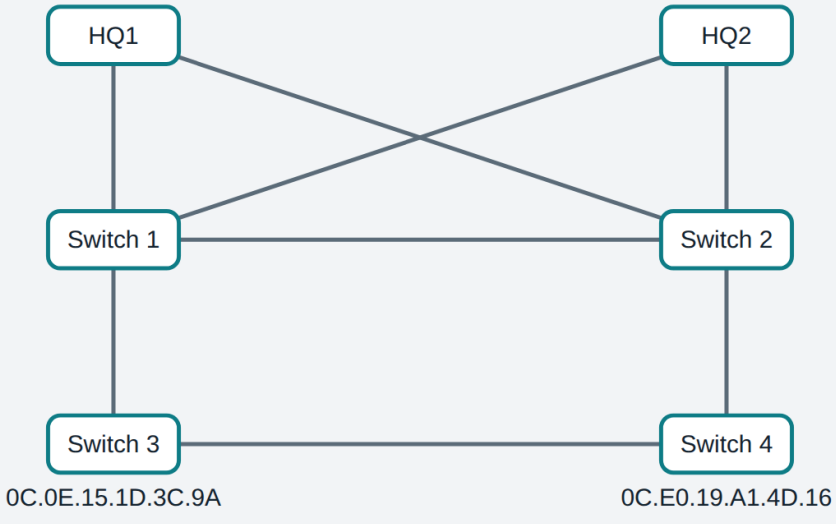
Source explanation

IEEE 802.1Q is the networking standard that supports virtual LANs (VLANs) on an Ethernet network. When a frame enters the VLAN-aware portion of the network (a trunk link, for example), a VLAN ID tag is added to represent the VLAN membership of that frame. The picture below shows how VLAN tag is added and removed while going through the network.

ccna-08-mi-q001 | STP & VTP Questions

Provided HTML: CCNA 200-301 Questions.html | Question 1

Refer to the exhibit. After the election process what is the root bridge in the HQ LAN?




```
Switch1: 0C.E0.38.57.24.22
Switch2: 0C.0E.15.22.1A.61
Switch3: 0C.0E.15.1D.3C.9A
Switch4: 0C.E0.19.A1.4D.16
```

A. Switch 2

B. Switch 1

C. Switch 4

D. Switch 3

Source answers: D

ccna-08-mi-q002 | STP & VTP Questions

Provided HTML: CCNA 200-301 Questions.html | Question 2

Which spanning-tree enhancement avoids the learning and listening states and immediately places ports in the forwarding state?

A. BPDUfilter

B. Backbonefast

C. BPDUguard

D. PortFast

Source answers: D

ccna-08-mi-q003 | STP & VTP Questions

Provided HTML: CCNA 200-301 Questions.html | Question 3

Which result occurs when PortFast is enabled on an interface that is connected to another switch?

A. VTP is allowed to propagate VLAN configuration information from switch to switch automatically.

B. Spanning tree may fail to detect a switching loop in the network that causes broadcast storms

C. Root port choice and spanning tree recalculation are accelerated when a switch link goes down

D. After spanning tree converges PortFast shuts down any port that receives BPDUs.

Source answers: B

Source explanation

Enabling the PortFast feature causes a switch or a trunk port to enter the STP forwarding-state immediately or upon a linkup event, thus bypassing the listening and learning states.
Note: To enable portfast on a trunk port you need the trunk keyword “spanning-tree portfast trunk“

ccna-08-mi-q004 | STP & VTP Questions

Provided HTML: CCNA 200-301 Questions.html | Question 4

Refer to exhibit. How does SW2 interact with other switches in this VTP domain?

```
SW2
vtp domain cisco
vtp mode transparent
vtp password test
interface fastethernet0/1
description connection to SW1
switchport mode trunk
switchport trunk encapsulation dot1q
```

A. It processes VTP updates from any VTP clients on the network on its access ports

B. It receives updates from all VTP servers and forwards all locally configured VLANs out all trunk ports

C. It transmits and processes VTP updates from any VTP Clients on the network on its trunk ports

D. It forwards only the VTP advertisements that it receives on its trunk ports

Source answers: D

Source explanation

The VTP mode of SW2 is transparent so it only forwards the VTP updates it receives to its trunk links without processing them.

ccna-08-mi-q005 | STP & VTP Questions

Provided HTML: CCNA 200-301 Questions.html | Question 5

Refer to the exhibit. What two conclusions should be made about this configuration? (Choose two)

```
SW1#show spanning-tree vlan 30

VLAN0030
Spanning tree enabled protocol rstp
Root ID  Priority  32798
         Address   0018.7363.4300
         Cost      2
         Port      13 (FastEthernet2/1)
         Hello Time 2 sec  Max Age 20 sec  Forward Delay 20 sec
<output omitted>
```

A. The spanning-tree mode is PVST+

B. The spanning-tree mode is Rapid PVST+

C. The root port is FastEthernet 2/1

D. This is a root bridge

E. The designated port is FastEthernet 2/1

Source answers: B, C

Source explanation

This bridge is not the root bridge because it does not have the statement “This bridge is the root”. When the local switch is not the root bridge, the port it shows would be the root port to the root bridge. Therefore in this case FastEthernet2/1 is the root port that is connected to the root bridge.

ccna-08-mi-q006 | STP & VTP Questions

Provided HTML: CCNA 200-301 Questions.html | Question 6

How does STP prevent forwarding loops at OSI Layer 2?

A. MAC address forwarding

B. Port blocking

C. Collision avoidance.

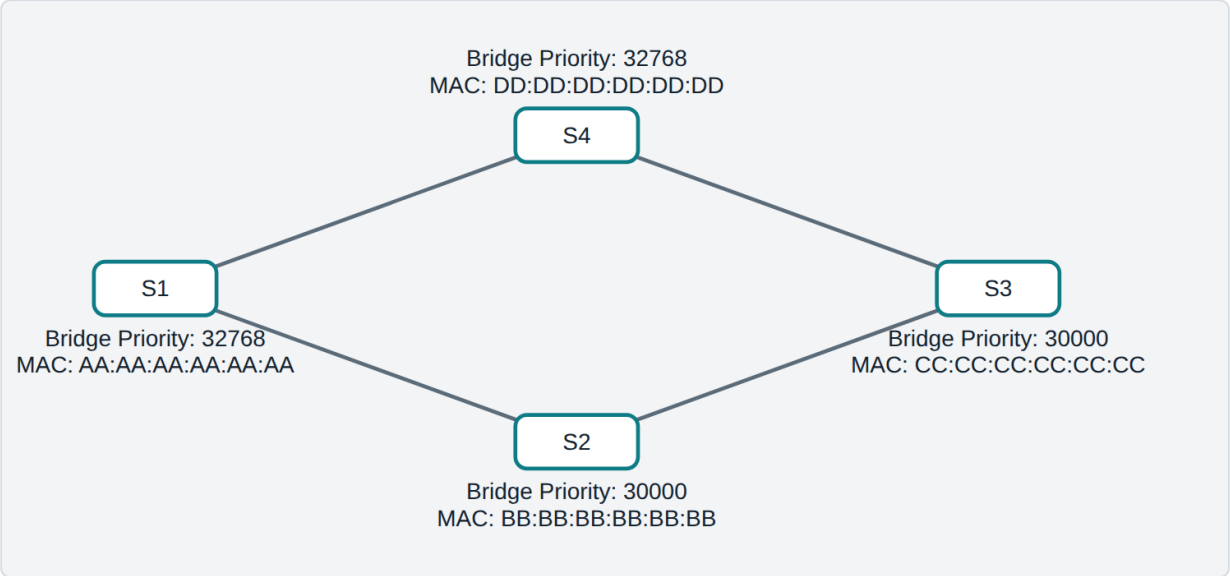
D. TTL

Source answers: B

ccna-08-mi-q007 | STP & VTP Questions

Provided HTML: CCNA 200-301 Questions.html | Question 7

Refer to the exhibit. Which switch becomes the root bridge?



A. S3

B. S1

C. S2

D. S4

Source answers: C

ccna-08-mi-q008 | STP & VTP Questions

Provided HTML: CCNA 200-301 Questions.html | Question 8

Which state does the switch port move to when PortFast is enabled?

A. learning

B. listening

C. forwarding

D. blocking

Source answers: C

ccna-08-mi-q009 | STP & VTP Questions

Provided HTML: CCNA 200-301 Questions.html | Question 9

Refer to the exhibit. Which switch in this configuration will be elected as the root bridge?




```
SW1: 0C:E0:38:00:36:75
SW2: 0C:0E:15:22:05:97
SW3: 0C:0E:15:1A:3C:9D
SW4: 0C:E0:18:A1:B3:19
```

A. SW4

B. SW2

C. SW1

D. SW3

Source answers: D

Source explanation

As this question did not state about bridge priority so we can assume all of them are using the default priority. Therefore the switch with lowest MAC address wins the election.

ccna-08-mi-q010 | STP & VTP Questions

Provided HTML: CCNA 200-301 Questions.html | Question 10

What criteria is used first during the root port selection process?

A. lowest neighbor’s bridge ID

B. lowest path cost to the root bridge

C. lowest neighbor’s port ID

D. local port ID

Source answers: B

ccna-08-mi-q011 | STP & VTP Questions

Provided HTML: CCNA 200-301 Questions.html | Question 11

When using Rapid PVST+, which command guarantees the switch is always the root bridge for VLAN 200?

A. spanning-tree vlan 200 root primary

B. spanning-tree vlan 200 priority 38572422

C. spanning-tree vlan 200 priority 614440

D. spanning-tree vlan 200 priority 0

Source answers: D

ccna-08-mi-q012 | STP & VTP Questions

Provided HTML: CCNA 200-301 Questions.html | Question 12

What is the primary effect of the spanning-tree portfast command?

A. It enables BPDU messages

B. It immediately puts the port into the forwarding state when the switch is reloaded

C. It minimizes spanning-tree convergence time

D. It immediately enables the port in the listening state

Source answers: C

Source explanation

The purpose of Port Fast is to minimize the time interfaces must wait for spanning-tree to converge, it is effective only when used on interfaces connected to end stations.

ccna-08-mi-q013 | STP & VTP Questions

Provided HTML: CCNA 200-301 Questions.html | Question 13

Which configuration ensures that the switch is always the root for VLAN 750?

A. Switch(config)#spanning-tree vlan 750 priority 38003685

B. Switch(config)#spanning-tree vlan 750 priority 0

C. Switch(config)#spanning-tree vlan 750 root primary

D. Switch(config)#spanning-tree vlan 750 priority 614440

Source answers: B

Source explanation

The keyword here is "always" so we should set the priority to the lowest value which is 0. If we use the "primary root" keyword then our switch is only set to the lower value compared to other switches. And new switch added in the future can be set to lower value than our switch to take over the root role.

ccna-10-mi-q001 | TCP & UDP Questions

Provided HTML: CCNA 200-301 Questions.html | Question 1

What is the difference regarding reliability and communication type between TCP and UDP?

A. TCP is not reliable and is a connection-oriented protocol; UDP is reliable and is a connectionless protocol

B. TCP is not reliable and is a connectionless protocol; UDP is reliable and is a connection-oriented protocol

C. TCP is reliable and is a connectionless protocol; UDP is not reliable and is a connection-oriented protocol

D. TCP is reliable and is a connection-oriented protocol UDP is not reliable and is a connectionless protocol

Source answers: D

ccna-10-mi-q002 | TCP & UDP Questions

Provided HTML: CCNA 200-301 Questions.html | Question 2

How do TCP and UDP differ in the way they provide reliability for delivery of packets?

A. TCP does not guarantee delivery or error checking to ensure that there is no corruption of data UDP provides message acknowledgement and retransmits data if lost

B. TCP provides flow control to avoid overwhelming a receiver by sending too many packets at once, UDP sends packets to the receiver in a continuous stream without checking for sequencing

C. TCP uses windowing to deliver packets reliably; UDP provides reliable message transfer between hosts by establishing a three-way handshake

D. TCP is a connectionless protocol that does not provide reliable delivery of data, UDP is a connection-oriented protocol that uses sequencing to provide reliable delivery

Source answers: B

ccna-10-mi-q003 | TCP & UDP Questions

Provided HTML: CCNA 200-301 Questions.html | Question 3

How do TCP and UDP differ in the way they guarantee packet delivery?

A. TCP uses checksum, parity checks, and retransmissions, and UDP uses acknowledgements only.

B. TCP uses two-dimensional parity checks, checksums, and cyclic redundancy checks and UDP uses retransmissions only.

C. TCP uses checksum, acknowledgement, and retransmissions, and UDP uses checksums only.

D. TCP uses retransmissions, acknowledgement and parity checks and UDP uses cyclic redundancy checks only.

Source answers: C

ccna-10-mi-q004 | TCP & UDP Questions

Provided HTML: CCNA 200-301 Questions.html | Question 4

How do TCP and UDP differ in the way that they establish a connection between two endpoints?

A. UDP provides reliable message transfer and TCP is a connectionless protocol

B. UDP uses SYN, SYN ACK and FIN bits in the frame header while TCP uses SYN, SYN ACK and ACK bits

C. TCP uses the three-way handshake and UDP does not guarantee message delivery

D. TCP uses synchronization packets, and UDP uses acknowledgment packets

Source answers: C

ccna-11-mi-q001 | IP Address & Subnetting Questions

Provided HTML: CCNA 200-301 Questions.html | Question 1

Refer to exhibit. Which statement explains the configuration error message that is received?

```
Router(config)#interface GigabitEthernet 1/0/1
Router(config-if)#ip address 192.168.16.143 255.255.255.240
Bad mask /28 for address 192.168.16.143
```

A. It belongs to a private IP address range

B. It is a network IP address

C. It is a broadcast IP address

D. The router does not support /28 mask

Source answers: C

ccna-11-mi-q002 | IP Address & Subnetting Questions

Provided HTML: CCNA 200-301 Questions.html | Question 2

Which network allows devices to communicate without the need to access the Internet?

A. 192.0.0.0/8

B. 209.165.201.0/24

C. 172.28.0.0/16

D. 172.9.0.0/16

Source answers: C

Source explanation

This question asks about the private ranges of IPv4 addresses. The private ranges of each class of IPv4 addresses:

Class A private IP address ranges from 10.0.0.0 to 10.255.255.255
Class B private IP address ranges from 172.16.0.0 to 172.31.255.255
Class C private IP address ranges from 192.168.0.0 to 192.168.255.255

Only the network 172.28.0.0/16 belongs to the private IP address (of class B).

ccna-11-mi-q003 | IP Address & Subnetting Questions

Provided HTML: CCNA 200-301 Questions.html | Question 3

An office has 8 floors with approximately 30-40 users per floor. What command must be configured on the router Switched Virtual Interface to use address space efficiently?

A. ip address 192.168.0.0 255.255.254.0

B. ip address 192.168.0.0 255.255.255.128

C. ip address 192.168.0.0 255.255.255.224

D. ip address 192.168.0.0 255.255.0.0

Source answers: C

Source explanation

We cannot use the subnet mask of 255.255.254.0 (/23) because it is smaller than the default subnet mask of class C (/24). If we use /23 subnet mask, in fact we are summarizing (supernet) it -> Answer "ip address 192.168.0.0 255.255.254.0" is not correct.

With the subnet mask of 255.255.255.128, we have 2^1 = 2 subnets only, not enough for 8 floors -> Answer "ip address 192.168.0.0 255.255.255.128" is not correct.

We need 40 users per subnet so our subnet must support at least 64 (=2^6) hosts (in fact 62 hosts). So the last octet of subnet mask should be 1100 0000 (with 6 bits “0”) = 192 -> The suitable subnet mask is 255.255.255.192. But this subnet mask only provides 2^2 = 4 subnets, which is not enough for 8 floors.

For 8 floors we need 1110 0000 (2^3 = 8) for the last octet of subnet mask so the subnet mask should be 255.255.255.224.

But with the subnet mask of 255.255.255.224, the host addresses per subnet is 2^5 - 2 = 30 so we cannot fulfill the host addresses for 40 users. We can only provide IPs for 30 users. This question said "approximately 30-40 users per floor" so "30 users per floor" is acceptable.

With “192.168.0.0 255.255.255.224” we have 8 subnets:
+ First subnet: 192.168.0.0 to 192.168.0.31
+ Second subnet: 192.168.0.32 to 192.168.0.63
+ Third subnet: 192.168.0.64 to 192.168.0.95
+ Fourth subnet: 192.168.0.96 to 192.168.0.127
+ Fifth subnet: 192.168.0.128 to 192.168.0.159
+ Sixth subnet: 192.168.0.160 to 192.168.0.191
+ Seventh subnet: 192.168.0.192 to 192.168.0.223
+ Eighth subnet: 192.168.0.224 to 192.168.0.255

ccna-11-mi-q004 | IP Address & Subnetting Questions

Provided HTML: CCNA 200-301 Questions.html | Question 4

Which function does the range of private IPv4 addresses perform?

A. provides a direct connection for hosts from outside of the enterprise network

B. allows multiple companies to each use the same addresses without conflicts

C. ensures that NAT is not required to reach the internet with private range addressing

D. enables secure communications to the internet for all external hosts

Source answers: B

ccna-11-mi-q005 | IP Address & Subnetting Questions

Provided HTML: CCNA 200-301 Questions.html | Question 5

A corporate office uses four floors in a building
* Floor 1 has 24 users
* Floor 2 has 29 users
* Floor 3 has 28 users
* Floor 4 has 22 users
Which subnet summarizes and gives the most efficient distribution of IP addresses for the router configuration?

A. 192.168.0.0/23 as summary and 192.168.0.0/25 for each floor

B. 192.168.0.0/25 as summary and 192.168.0.0/27 for each floor

C. 192.168.0.0/26 as summary and 192.168.0.0/29 for each floor

D. 192.168.0.0/24 as summary and 192.168.0.0/28 for each floor

Source answers: B

Source explanation

We see the maximum number of user per floor is 29 users (Floor 2) < 32 so the best subnet mask should be 1110 0000 which allows 2^5 – 2 = 30 hosts per subnet.

ccna-11-mi-q006 | IP Address & Subnetting Questions

Provided HTML: CCNA 200-301 Questions.html | Question 6

Which goal is achieved by the implementation of private IPv4 addressing on a network?

A. provides a reduction in size of the forwarding table on network routers

B. provides an added level of protection against Internet exposure

C. allows servers and workstations to communicate across public network boundaries

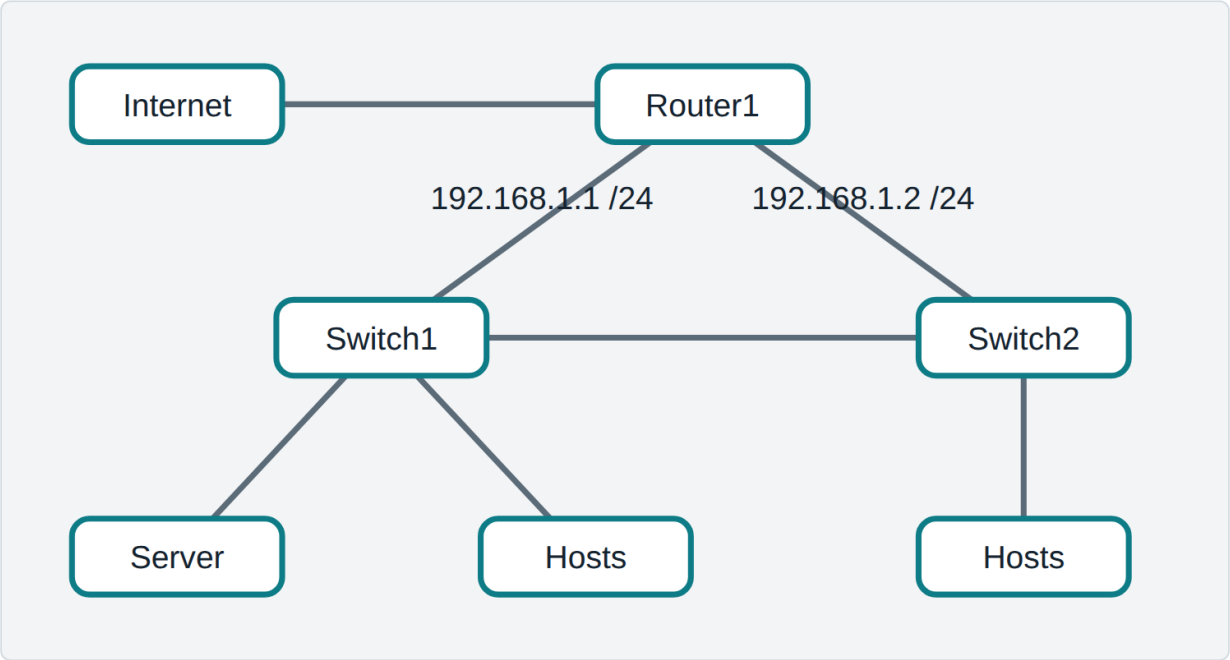
D. allows communication across the Internet to other private networks

Source answers: B

ccna-11-mi-q007 | IP Address & Subnetting Questions

Provided HTML: CCNA 200-301 Questions.html | Question 7

Refer to the exhibit. A network technician is asked to design a small network with redundancy. The exhibit represents this design, with all hosts configured in the same VLAN. What conclusions can be made about this design?



A. The design will function as intended

B. The connection between switches should be a trunk.

C. The router interfaces must be encapsulated with the 802.1Q protocol.

D. Spanning-tree will need to be used.

E. The router will not accept the addressing scheme.

Source answers: E

Source explanation

Each interface on a router must be in a different network. If two interfaces are in the same network, the router will not accept it and show error when the administrator assigns it.

ccna-11-mi-q008 | IP Address & Subnetting Questions

Provided HTML: CCNA 200-301 Questions.html | Question 8

An engineer must configure a /30 subnet between two routers. Which usable IP address and subnet mask combination meets this criteria?

A. interface e0/0
description to HQ-A371:10975
ip address 10.2.1.3 255.255.255.252

B. interface e0/0
description to HQ-A371:10975
ip address 172.16.1.4 255.255.255.248

C. interface e0/0
description to HQ-A371:10975
ip address 192.168.1.1 255.255.255.248

D. interface e0/0
description to HQ-A371:10975
ip address 209.165.201.2 255.255.255.252

Source answers: D

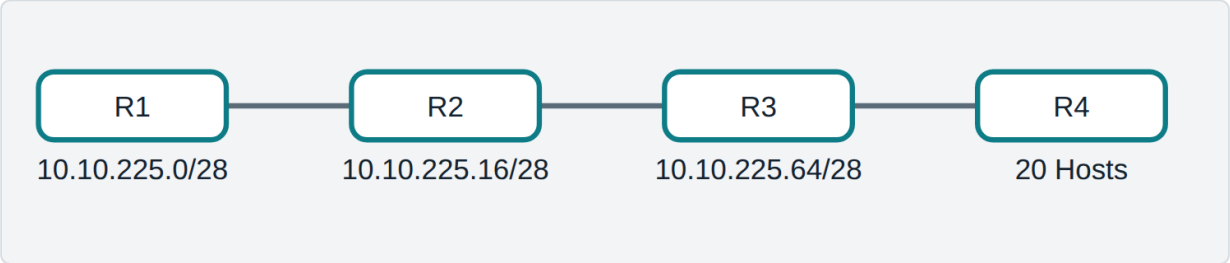
Source explanation

A /30 subnet means subnet mask of 255.255.255.252. But 10.2.1.3 255.255.255.252 is a broadcast IP address; only 209.165.201.2/30 is the usable IP address.

ccna-11-mi-q009 | IP Address & Subnetting Questions

Provided HTML: CCNA 200-301 Questions.html | Question 9

Refer to the exhibit. An engineer must add a subnet for a new office that will add 20 users to the network. Which IPv4 network and subnet mask combination does the engineer assign to minimize wasting addresses?



A. 10.10.225.48 255.255.255.224

B. 10.10.225.32 255.255.255.224

C. 10.10.225.48 255.255.255.240

D. 10.10.225.32 255.255.255.240

Source answers: B

Source explanation

We need a subnet with 20 users so we need 5 bits 0 in the subnet mask as 2^5 – 2 = 30 > 20. Therefore the subnet mask should be /27 (with last octet is 1110 0000 in binary). The increment is 32 so the valid network address is 10.10.225.32.

ccna-13-mi-q001 | IP Routing Questions 2

Provided HTML: CCNA 200-301 Questions.html | Question 1

Refer to the exhibit. Router R1 Fa0/0 cannot ping router R3 Fa0/1. Which action must be taken in router R1 to help resolve the configuration issue?

```
R1 Fa0/0 10.10.10.1/24 --- Fa0/0 10.10.10.2/24 R2 Fa0/1 20.20.20.1/24 --- Fa0/1
20.20.20.2/24 R3

R1#show ip route
<output omitted>
Gateway of last resort is not set
      10.0.0.0/24 is subnetted, 1 subnets
C        10.10.10.0 is directly connected, FastEthernet0/0

R2#show ip route
<output omitted>
Gateway of last resort is not set
      20.0.0.0/24 is subnetted, 1 subnets
C        20.20.20.0 is directly connected, FastEthernet0/1
      10.0.0.0/24 is subnetted, 1 subnets
C        10.10.10.0 is directly connected, FastEthernet0/0

R3#show ip route
<output omitted>
Gateway of last resort is not set
      20.0.0.0/24 is subnetted, 1 subnets
C        20.20.20.0 is directly connected, FastEthernet0/1
      10.0.0.0/24 is subnetted, 1 subnets
S        10.10.10.0 [1/0] via 20.20.20.1
```

A. configure a static route with Fa0/1 as the egress interface to reach the 20.20.20.0/24 network

B. configure a static route with 10.10.10.2 as the next hop to reach the 20.20.20.0/24 network

C. set the default gateway as 20.20.20.2

D. set the default network as 20.20.20.0/24

Source answers: B

ccna-13-mi-q002 | IP Routing Questions 2

Provided HTML: CCNA 200-301 Questions.html | Question 2

R1 has learned route 10.10.10.0/24 via numerous routing protocols. Which route is installed?

A. route with the next hop that has the highest IP

B. route with the lowest administrative distance

C. route with the lowest cost

D. route with the shortest prefix length

Source answers: B

ccna-13-mi-q003 | IP Routing Questions 2

Provided HTML: CCNA 200-301 Questions.html | Question 3

Router R1 must send all traffic without a matching routing-table entry to 192.168.1.1. Which configuration accomplishes this task?

A. R1#config t
R1(config)#ip routing
R1(config)#ip route 0.0.0.0 0.0.0.0 192.168.1.1

B. R1#config t
R1(config)#ip routing
R1(config)#ip default-gateway 192.168.1.1

C. R1#config t
R1(config)#ip routing
R1(config)#ip route 192.168.1.1 0.0.0.0 0.0.0.0

D. R1#config t
R1(config)#ip routing
R1(config)#ip route default-route 192.168.1.1

Source answers: A

ccna-13-mi-q004 | IP Routing Questions 2

Provided HTML: CCNA 200-301 Questions.html | Question 4

Refer to the exhibit. Router R1 is running three different routing protocols. Which route characteristic is used by the router to forward the packet that it receives for destination IP 172.16.32.1?

```
R1#show ip route
.....
D 172.16.32.0/27 [90/293847483] via 20.1.1.1
O 172.16.32.0/19 [110/22847] via 20.1.1.10
R 172.16.32.0/24 [120/2] via 20.1.1.3
```

A. longest prefix

B. administrative distance

C. cost

D. metric

Source answers: A

Source explanation

The router evaluates routes in the following order:
1. Prefix Length - The longest-matching route is preferred first. Prefix length trumps all other route attributes.
2. Administrative Distance - In the event there are multiple routes to a destination with the same prefix length, the route learned by the protocol with the lowest administrative distance is preferred.
3. Metric - In the event there are multiple routes learned by the same protocol with same prefix length, the route with the lowest metric is preferred. (If two or more of these routes have equal metrics, load balancing across them may occur.)

ccna-13-mi-q005 | IP Routing Questions 2

Provided HTML: CCNA 200-301 Questions.html | Question 5

A packet is destined for 10.10.1.22. Which static route does the router choose to forward the packet?

A. ip route 10.10.1.0 255.255.255.240 10.10.255.1

B. ip route 10.10.1.20 255.255.255.252 10.10.255.1

C. ip route 10.10.1.20 255.255.255.254 10.10.255.1

D. ip route 10.10.1.16 255.255.255.252 10.10.255.1

Source answers: B

Source explanation

The subnet 10.10.1.20/30 covers the destination 10.10.1.22 as this subnet ranges from 10.10.1.20 to 10.10.1.23 so it is the correct answer.

ccna-13-mi-q006 | IP Routing Questions 2

Provided HTML: CCNA 200-301 Questions.html | Question 6

Refer to the exhibit. Router R2 is configured with multiple routes to reach network 10.1.1.0/24 from router R1. What protocol is chosen by router R2 to reach the destination network 10.1.1.0/24?

```
R1(config)#eigrp router-id 1.1.1.1
R1(config)#network 10.1.1.0 0.0.0.255
R1(config)#network 192.168.0.1 0.0.0.0

R2(config)#interface gi1/1
R2(config-if)#ip address 192.168.0.2 255.255.255.0

R2(config)#router bgp 65001
R2(config-router)#neighbor 192.168.0.1 remote-as 65000

R2(config)#router ospf 1
R2(config)#router-id 2.2.2.2
R2(config)#network 192.168.1.2 0.0.0.0 area 0

R2(config)#router eigrp 1
R2(config)#eigrp router-id 1.1.1.1
R2(config)#network 192.168.0.1 0.0.0.0

R2(config)#ip route 10.1.1.0 255.255.255.0 192.168.0.1
```

A. eBGP

B. EIGRP

C. static

D. OSPF

Source answers: C

Source explanation

The static route has the Administrative Distance of 1 so it is preferred over other popular routing protocols.

ccna-13-mi-q007 | IP Routing Questions 2

Provided HTML: CCNA 200-301 Questions.html | Question 7

Refer to the exhibit. How does router R1 handle traffic to 192.168.12.16?

```
EIGRP: 192.168.12.0/24
RIP: 192.168.12.0/27
OSPF: 192.168.12.0/26
```

A. It selects the OSPF route because it has the lowest cost

B. It selects the EIGRP route because it has the lowest administrative distance

C. It selects the IS-IS route because it has the shortest prefix inclusive of the destination address

D. It selects the RIP route because it has the longest prefix inclusive of the destination address

Source answers: D

ccna-13-mi-q008 | IP Routing Questions 2

Provided HTML: CCNA 200-301 Questions.html | Question 8

Refer to the exhibit. A router reserved these five routes from different routing information sources. Which two routes does the router install in its routing table? (Choose two)

```
IBGP route 10.0.0.0/30
RIP route 10.0.0.0/30
OSPF route 10.0.0.0/16
OSPF route 10.0.0.0/30
EIGRP route 10.0.0.1/32
```

A. OSPF route 10.0.0.0/16

B. RIP route 10.0.0.0/30

C. OSPF route 10.0.0.0/30

D. EIGRP route 10.0.0.1/32

E. iBGP route 10.0.0.0/30

Source answers: C, D

Source explanation

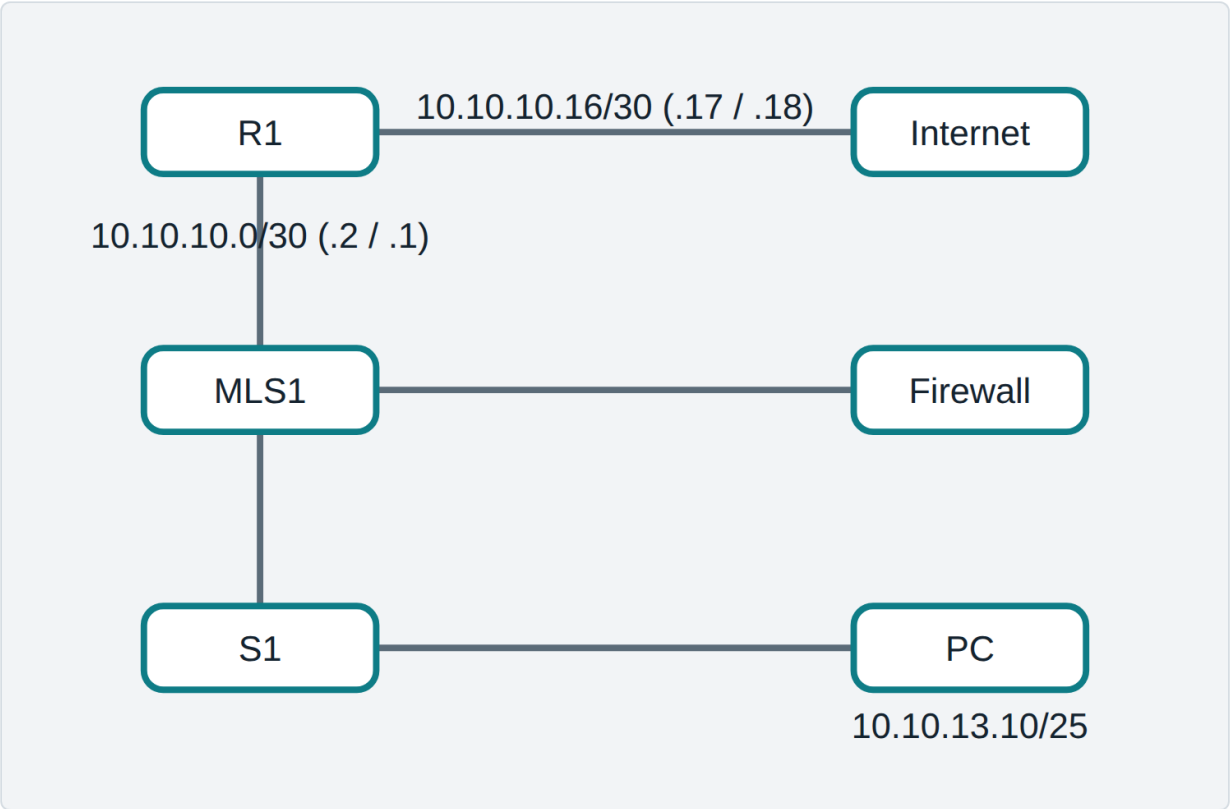
In the five above routes, there are only three different routes which are 10.0.0.0/30 and 10.0.0.0/16 and 10.0.0.1/32. Since each of these routes has a different prefix length (subnet mask), they’re considered different destinations, and they will all be installed in the routing table.

The prefix 10.0.0.0/30 was learned from three different routing protocols IBGP, RIP and OSPF so the route with lowest AD would be chosen. The AD of IBGP is 200 and the AD of RIP is 120 so for the prefix 10.0.0.0/30, only the OSPF route has the lowest AD (110) and it would be chosen to install into the routing table.

ccna-13-mi-q009 | IP Routing Questions 2

Provided HTML: CCNA 200-301 Questions.html | Question 9

Refer to the exhibit. Which route type is configured to reach the internet?




```
R1#show ip route
Gateway of last resort is 10.10.10.18 to network 0.0.0.0
      10.0.0.0/8 is variably subnetted, 4 subnets, 3 masks
C        10.10.10.0/30 is directly connected, FastEthernet0/1
O        10.10.13.0/25 [110/4576] via 10.10.10.1, 02:53:11, FastEthernet0/1
C        10.10.10.16/30 is directly connected, FastEthernet0/24
O        10.10.13.144/28 [110/110] via 10.10.10.1, 03:51:21, FastEthernet0/1
B*    0.0.0.0/0 [20/0] via 10.10.10.18, 02:17:53
```

A. floating static route

B. host route

C. network route

D. default route

Source answers: D

Source explanation

R1 uses the last entry (B* 0.0.0.0/0 ...) to reach the Internet. The symbol * means this is a default route.

ccna-13-mi-q010 | IP Routing Questions 2

Provided HTML: CCNA 200-301 Questions.html | Question 10

What are two reasons for an engineer to configure a floating static route? (Choose two)

A. to support load balancing via static routing

B. to control the return path of traffic that is sent from the router

C. to automatically route traffic on a secondary path when the primary path goes down

D. to route traffic differently based on the source IP of the packet

E. to enable fallback static routing when the dynamic routing protocol fails

Source answers: C, E

Source explanation

Floating static routes are static routes that have an administrative distance greater than the administrative distance (AD) of another static route or dynamic routes. By default a static route has an AD of 1 then floating static route must have the AD greater than 1. Floating static route has a manually configured administrative distance greater than that of the primary route and therefore would not be in the routing table until the primary route fails.

ccna-13-mi-q011 | IP Routing Questions 2

Provided HTML: CCNA 200-301 Questions.html | Question 11

Refer to the exhibit. A packet is being sent across router R1 to host 172.16.3.14. To which destination does the router send the packet?

```
R1# show ip route | begin gateway
Gateway of last resort is 209.165.200.246 to network 0.0.0.0
S*    0.0.0.0/0 [1/0] via 209.165.200.246, Serial0/1/0
                  is directly connected, Serial0/1/0
      172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
S        172.16.3.0/24 [1/0] via 207.165.200.250, Serial0/0/0
O        172.16.3.0/28 [110/84437] via 207.165.200.254, 00:00:30, Serial0/0/1
      207.165.200.0/24 is variably subnetted, 6 subnets, 2 masks
C        207.165.200.244/30 is directly connected, Serial0/1/0
L        207.165.200.245/32 is directly connected, Serial0/1/0
C        207.165.200.248/30 is directly connected, Serial0/0/0
L        207.165.200.249/32 is directly connected, Serial0/0/0
C        207.165.200.252/30 is directly connected, Serial0/0/1
L        207.165.200.253/32 is directly connected, Serial0/0/1
```

A. 207.165.200.254 via Serial0/0/1

B. 207.165.200.250 via Serial0/0/0

C. 207.165.200.254 via Serial0/0/0

D. 207.165.200.246 via Serial0/1/0

Source answers: A

ccna-13-mi-q012 | IP Routing Questions 2

Provided HTML: CCNA 200-301 Questions.html | Question 12

Refer to the exhibit. Which command configures a floating static route to provide a backup to the primary link?

```
R1#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
Gateway of last resort is 209.165.202.131 to network 0.0.0.0
S*    0.0.0.0/0 [1/0] via 209.165.202.131
      209.165.200.0/27 is subnetted, 1 subnets
S        209.165.200.224 [254/0] via 209.165.202.129
      209.165.201.0/27 is subnetted, 1 subnets
S        209.165.201.0 [1/0] via 209.165.202.130
```

A. ip route 209.165.200.224 255.255.255.224 209.165.202.129 254

B. ip route 0.0.0.0 0.0.0.0 209.165.200.224

C. ip route 0.0.0.0 0.0.0.0 209.165.202.131

D. ip route 209.165.201.0 255.255.255.224 209.165.202.130

Source answers: A

Source explanation

Floating static routes are static routes that have an administrative distance greater than the administrative distance (AD) of another static route or dynamic routes. By default a static route has an AD of 1 then floating static route must have the AD greater than 1. Floating static route has a manually configured administrative distance greater than that of the primary route and therefore would not be in the routing table until the primary route fails.

In this question, the floating static route to subnet 209.165.200.224/27 is manually assigned the AD of 254.

ccna-14-mi-q001 | OSPF Questions

Provided HTML: CCNA 200-301 Questions.html | Question 1

You have configured a router with an OSPF router ID, but its IP address still reflects the physical interface. Which action can you take to correct the problem in the least disruptive way?

A. Save the router configuration

B. Reload the router

C. Reload the OSPF process

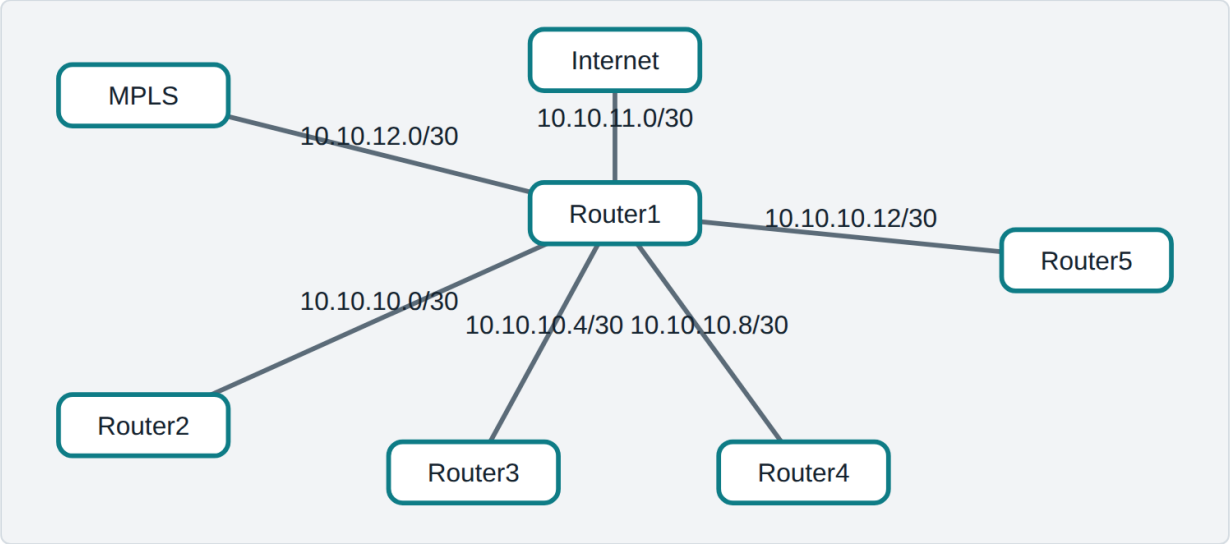
D. Specify a loopback address

Source answers: C

ccna-14-mi-q002 | OSPF Questions

Provided HTML: CCNA 200-301 Questions.html | Question 2

Refer to the exhibit. To which device does Router1 send packets that are destined to host 10.10.13.165?




```
Router1#show ip route
Gateway of last resort is 10.10.11.2 to network 0.0.0.0
      209.165.200.0/27 is subnetted, 1 subnets
B        209.165.200.224 [20/0] via 10.10.12.2,03:32:14
      209.165.201.0/27 is subnetted, 1 subnets
B        209.165.201.0 [20/0] via 10.10.12.2,02:26:53
      209.165.202.0/27 is subnetted, 1 subnets
B        209.165.202.128 [20/0] via 10.10.12.2,02:46:03
      10.0.0.0/8 is variably subnetted, 10 subnets, 4 masks
O        10.10.13.0/25 [110/2] via 10.10.10.1,00:00:04, GigabitEthernet0/0
O        10.10.13.128/28 [110/2] via 10.10.10.5,00:00:12, GigabitEthernet0/1
O        10.10.13.144/28 [110/2] via 10.10.10.9,00:01:57, GigabitEthernet0/2
O        10.10.13.160/29 [110/2] via 10.10.10.5,00:00:12, GigabitEthernet0/1
O        10.10.13.208/29 [110/2] via 10.10.10.13,00:01:57, GigabitEthernet0/3
S*    0.0.0.0/0 [1/0] via 10.10.11.2
```

A. Router2

B. Router5

C. Router3

D. Router4

Source answers: C

Source explanation

The destination of 10.10.13.165 matches the entry “O 10.10.13.160/29 [110/2]...” because of the longest prefix length rule so the packet will be forwarded to 10.10.10.5, which is Router3.

ccna-14-mi-q003 | OSPF Questions

Provided HTML: CCNA 200-301 Questions.html | Question 3

Refer to the exhibit. What does router R1 use as its OSPF router-ID?

```
R1#show ip interface brief
Interface            IP-Address     OK? Method Status                Protocol
FastEthernet0/0      unassigned     YES NVRAM  administratively down down
GigabitEthernet1/0   192.168.0.1    YES NVRAM  up                    up
GigabitEthernet2/0   10.10.1.10     YES NVRAM  up                    up
GigabitEthernet3/0   10.10.10.20    YES NVRAM  up                    up
GigabitEthernet4/0   unassigned     YES NVRAM  administratively down down
Loopback0            172.16.15.10   YES manual up                    up
```

A. 10.10.10.20

B. 192.168.0.1

C. 10.10.1.10

D. 172.16.15.10

Source answers: D

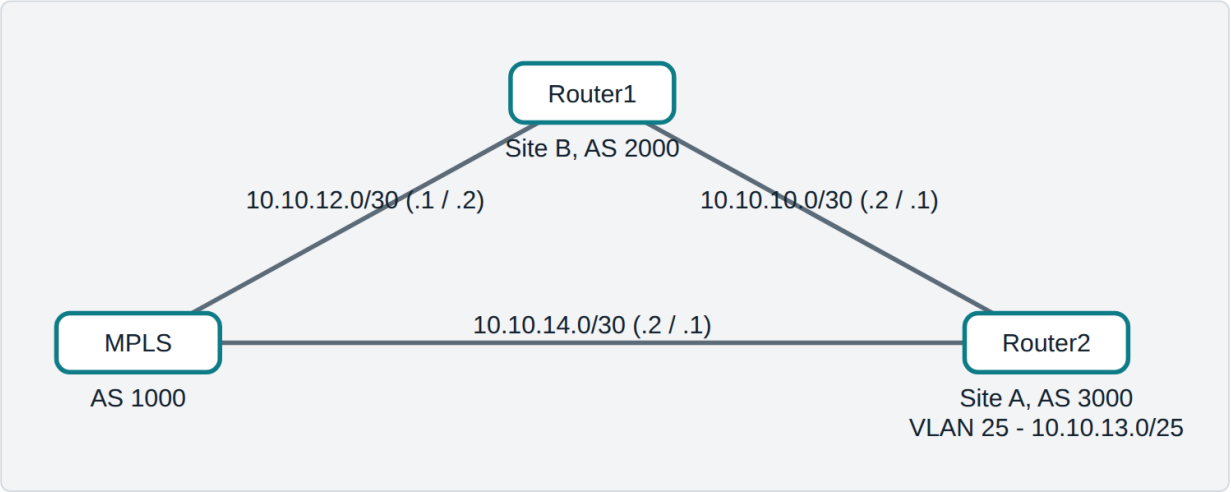
Source explanation

1. Manual configuration of the router ID (via the “router-id x.x.x.x” command under OSPF router configuration mode).
2. Highest IP address on a loopback interface.
3. Highest IP address on a non-loopback and active (no shutdown) interface.

ccna-14-mi-q004 | OSPF Questions

Provided HTML: CCNA 200-301 Questions.html | Question 4

Refer to the exhibit. An engineer is bringing up a new circuit to the MPLS provider on the Gi0/1 interface of Router1. The new circuit uses eBGP and learns the route to VLAN25 from the BGP path.
What is the expected behavior for the traffic flow for route 10.10.13.0/25?




```
Router1#show ip route
Gateway of last resort is 10.10.11.2 to network 0.0.0.0
      10.0.0.0/8 is variably subnetted, 8 subnets, 4 masks
C        10.10.10.0/28 is directly connected, GigabitEthernet0/0
C        10.10.11.0/30 is directly connected, FastEthernet2/0
O        10.10.13.0/25 [110/2] via 10.10.10.1,00:00:32, GigabitEthernet0/0
O        10.10.13.128/28 [110/2] via 10.10.10.1,00:00:32, GigabitEthernet0/0
O        10.10.13.144/28 [110/2] via 10.10.10.1,00:00:32, GigabitEthernet0/0
O        10.10.13.160/29 [110/2] via 10.10.10.1,00:00:32, GigabitEthernet0/0
O        10.10.13.208/30 [110/2] via 10.10.10.1,00:00:32, GigabitEthernet0/0
O        10.10.13.252/30 [110/2] via 10.10.10.1,00:00:32, GigabitEthernet0/0
S*    0.0.0.0/0 [1/0] via 10.10.11.2
```

A. Route 10.10.13.0/25 learned via the Gi0/0 interface remains in the routing table

B. Traffic to 10.10.13.0.25 is load balanced out of multiple interfaces

C. Traffic to 10.10.13.0/25 is symmetrical

D. Route 10.10.13.0/25 is updated in the routing table as being learned from interface Gi0/1

Source answers: D

Source explanation

From the output we can deduce interface Gi0/0 on R1 is the interface connecting to Router2 so interface Gi0/1 is the interface connecting to the new MPLS AS 1000.

The AD of eBGP (20) is smaller than that of OSPF (110) so the route to 10.10.13.0/25 will be updated as being learned from the new BGP path.

ccna-14-mi-q005 | OSPF Questions

Provided HTML: CCNA 200-301 Questions.html | Question 5

A user configured OSPF in a single area between two routers. A serial interface connecting R1 and R2 is running encapsulation PPP. By default which OSPF network type is seen on this interface when the user types show ip ospf interface on R1 or R2?

A. point-to-multipoint

B. non-broadcast

C. point-to-point

D. broadcast

Source answers: C

Source explanation

The default OSPF network type for HDLC and PPP on Serial link is point-to-point (while the default OSPF network type for Ethernet link is Broadcast).

ccna-14-mi-q006 | OSPF Questions

Provided HTML: CCNA 200-301 Questions.html | Question 6

Refer to the exhibit. If OSPF is running on this network, how does Router 2 handle traffic from Site B to 10.10.13.128/25 at Site A?

```
Site A -- Router1 ======== Router2 -- Site B

Router2#show ip route
Gateway of last resort is not set
      10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C        10.10.10.8/30 is directly connected, FastEthernet0/2
C        10.10.10.12/30 is directly connected, FastEthernet0/1
O        10.10.13.0/25 [110/11] via 10.10.10.9, 00:00:02, FastEthernet0/2
                       [110/11] via 10.10.10.13, 00:00:02, FastEthernet0/1
C        10.10.10.4/30 is directly connected, FastEthernet0/2
```

A. It sends packets out of interface Fa0/1 only

B. It is unreachable and discards the traffic

C. It load-balances traffic out of Fa0/1 and Fa0/2

D. It sends packets out of interface Fa0/2 only

Source answers: B

Source explanation

Router2 does not have an entry for the subnet 10.10.13.128/25. It only has an entry for 10.10.13.0/25, which ranges from 10.10.13.0 to 10.10.13.127.

ccna-14-mi-q007 | OSPF Questions

Provided HTML: CCNA 200-301 Questions.html | Question 7

Refer to the exhibit. The default-information originate command is configured under the R1 OSPF configuration. After testing, workstations on VLAN 20 at Site B cannot reach a DNS server on the Internet. Which action corrects the configuration issue?

```
Site B: R2 (.1) --10.10.10.0/30-- (.2) R1 :Site A (.17) --10.10.10.16/30-- (.18)
Internet

R2#show run | b router ospf
router ospf 1
 router-id 2.2.2.2
 log-adjacency-changes
 auto-cost reference-bandwidth 10000
 network 10.10.10.1 0.0.0.0 area 0
 network 10.10.13.1 0.0.0.0 area 0

R1#show run | b router ospf
router ospf 1
 router-id 1.1.1.1
 log-adjacency-changes
 auto-cost reference-bandwidth 10000
 network 10.10.10.2 0.0.0.0 area 0
 default-information originate

R2#show ip route
Gateway of last resort is not set
      10.0.0.0/8 is variably subnetted, 3 subnets, 3 masks
C        10.10.10.0/30 is directly connected
C        10.10.13.0/25 is directly connected, Vlan20
C        10.10.13.144/28 is directly connected, Vlan40

R1#show ip route
Gateway of last resort is not set
      10.0.0.0/8 is variably subnetted, 4 subnets, 3 masks
C        10.10.10.0/30 is directly connected, FastEthernet0/1
O        10.10.13.0/25 [110/5766] via 10.10.10.1,00:33:22
C        10.10.10.16/30 is directly connected, FastEthernet0/24
O        10.10.13.144/28 [110/110] via 10.10.10.1, 01:33:33
```

A. Configure the ip route 0.0.0.0 0.0.0.0 10.10.10.18 command on R1

B. Add the always keyword to the default-information originate command on R1

C. Configure the ip route 0.0.0.0 0.0.0.0 10.10.10.2 command on R2

D. Add the default-information originate command on R2

Source answers: A

Source explanation

The default-information originate command advertises a default route into a normal area, provided the advertising router already has a default route. But in the routing table of R1 we don’t see such a default route. Therefore we have to configure one.

We cannot only add the "always" keyword to the command "default-information originate" on R1 because R2 still does not know how to forward traffic received from R1.

ccna-14-mi-q008 | OSPF Questions

Provided HTML: CCNA 200-301 Questions.html | Question 8

An engineer configured an OSPF neighbor as a designated router. Which state verifies the designated router is in the proper mode?

A. Exchange

B. Full

C. 2-way

D. Init

Source answers: B

ccna-14-mi-q009 | OSPF Questions

Provided HTML: CCNA 200-301 Questions.html | Question 9

What is the purpose of the show ip ospf interface command?

A. displaying OSPF neighbor information on a per-interface-type basis

B. displaying general information about OSPF routing processes

C. displaying OSPF-related interface information

D. displaying OSPF neighbor information on a per-interface basis

Source answers: C

ccna-14-mi-q010 | OSPF Questions

Provided HTML: CCNA 200-301 Questions.html | Question 10

R1 has learned route 192.168.12.0/24 via IS-IS, OSPF, RIP and Internal EIGRP. Under normal operating conditions, which routing protocol is installed in the routing table?

A. RIP

B. IS-IS

C. OSPF

D. Internal EIGRP

Source answers: D

Source explanation

With the same route (prefix), the router will choose the routing protocol with lowest Administrative Distance (AD) to install into the routing table. The AD of Internal EIGRP (90) is lowest so it would be chosen. The table below lists the ADs of popular routing protocols.

| Routing protocol | AD

| Directly Connected | 0

| Static | 1

| EIGRP | 90

| EIGRP Summary route | 5

| OSPF | 110

| RIP | 120

Note: The AD of IS-IS is 115. The “EIGRP” in the table above is “Internal EIGRP”. The AD of “External EIGRP” is 170. An EIGRP external route is a route that was redistributed into EIGRP.

> With the same route (prefix), the router will choose the routing protocol with lowest Administrative Distance (AD) to install into the routing table. The AD of Internal EIGRP (90) is lowest so it would be chosen. The table below lists the ADs of popular routing protocols.

Routing protocol	AD
Directly Connected	0
Static	1
EIGRP	90
EIGRP Summary route	5
OSPF	110
RIP	120

> Note: The AD of IS-IS is 115. The “EIGRP” in the table above is “Internal EIGRP”. The AD of “External EIGRP” is 170. An EIGRP external route is a route that was redistributed into EIGRP.

ccna-14-mi-q011 | OSPF Questions

Provided HTML: CCNA 200-301 Questions.html | Question 11

Refer to the exhibit. The “show ip ospf interface” command has been executed on R1. How is OSPF configured?

```
Designated Router (ID) 10.11.11.11, Interface address 10.10.10.1
Backup Designated router (ID) 10.3.3.3, Interface address 10.10.10.3
Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
  oob-resync timeout 40
  Hello due in 00:00:08
Supports Link-local Signaling (LLS)
Cisco NSF helper support enabled
IETF NSF helper support enabled
...
Last flood scan time is 0 msec, maximum is 1 msec
Neighbor Count is 3, Adjacent neighbor count is 3
  Adjacent with neighbor 10.1.1.4
  Adjacent with neighbor 10.2.2.2
  Adjacent with neighbor 10.3.3.3 (Backup Designated Router)
Suppress hello for 0 neighbor(s)
```

A. The interface is not participating in OSPF

B. A point-to-point network type is configured

C. There are six OSPF neighbors on this interface

D. The default Hello and Dead timers are in use

Source answers: D

Source explanation

From the output we can see there are Designated Router & Backup Designated Router for this OSPF domain so this is a broadcast network (point-to-point and point-to-multipoint networks do not elect DR & BDR) -> Answer 'A point-to-point network type is configured' is not correct.

By default, the timers on a broadcast network (Ethernet, point-to-point and point-to-multipoint) are 10 seconds hello and 40 seconds dead (therefore answer 'The default Hello and Dead timers are in use' is correct). The timers on a non-broadcast network are 30 seconds hello 120 seconds dead.

From the line “Neighbor Count is 3”, we learn there are four OSPF routers in this OSPF domain -> Answer 'There are six OSPF neighbors on this interface' is not correct.

ccna-14-mi-q012 | OSPF Questions

Provided HTML: CCNA 200-301 Questions.html | Question 12

When OSPF learns multiple paths to a network, how does it select a route?

A. It multiple the active K value by 256 to calculate the route with the lowest metric.

B. For each existing interface, it adds the metric from the source router to the destination to calculate the route with the lowest bandwidth.

C. It divides a reference bandwidth of 100 Mbps by the actual bandwidth of the existing interface to calculate the router with the lowest cost.

D. It count the number of hops between the source router and the destination to determine the router with the lowest metric

Source answers: C

ccna-14-mi-q013 | OSPF Questions

Provided HTML: CCNA 200-301 Questions.html | Question 13

A user configured OSPF and advertised the Gigabit Ethernet interface in OSPF. By default, which type of OSPF network does this interface belong to?

A. nonbroadcast

B. broadcast

C. point-to-point

D. point-to-multipoint

Source answers: B

Source explanation

The Broadcast network type is the default for an OSPF enabled ethernet interface (while Point-to-Point is the default OSPF network type for Serial interface with HDLC and PPP encapsulation).

Reference: https://www.oreilly.com/library/view/cisco-ios-cookbook/0596527225/ch08s15.html

ccna-16-mi-q001 | EIGRP Questions

Provided HTML: CCNA 200-301 Questions.html | Question 1

Refer to the exhibit.
Which route type does the routing protocol Code D represent in the output?

```
      10.0.0.0/24 is subnetted, 1 subnets
C        10.0.0.0 is directly connected, FastEthernet0/1
C     172.16.0.0/16 is directly connected, FastEthernet0/0
D     192.168.0.0/24 [90/30720] via 172.16.0.2, 00:00:04, FastEthernet0/0
```

A. internal BGP route

B. statically assigned route

C. /24 route of a locally configured IP

D. route learned through EIGRP

Source answers: D

Source explanation

Code “D” represents for EIGRP. Code “E” could not be used for EIGRP as it was used for Exterior Gateway Protocol (EGP), which is BGP predecessor.

ccna-16-mi-q002 | EIGRP Questions

Provided HTML: CCNA 200-301 Questions.html | Question 2

A router running EIGRP has learned the same route from two different paths. Which parameter does the router use to select the best path?

A. administrative distance

B. metric

C. as-path

D. cost

Source answers: B

ccna-16-mi-q003 | EIGRP Questions

Provided HTML: CCNA 200-301 Questions.html | Question 3

By default, how does EIGRP determine the metric of a route for the routing table?

A. It uses a default metric of 10 for all routes that are learned by the router

B. It counts the number of hops between the receiving and destination routers and uses that value as the metric

C. It uses the bandwidth and delay values of the path to calculate the route metric

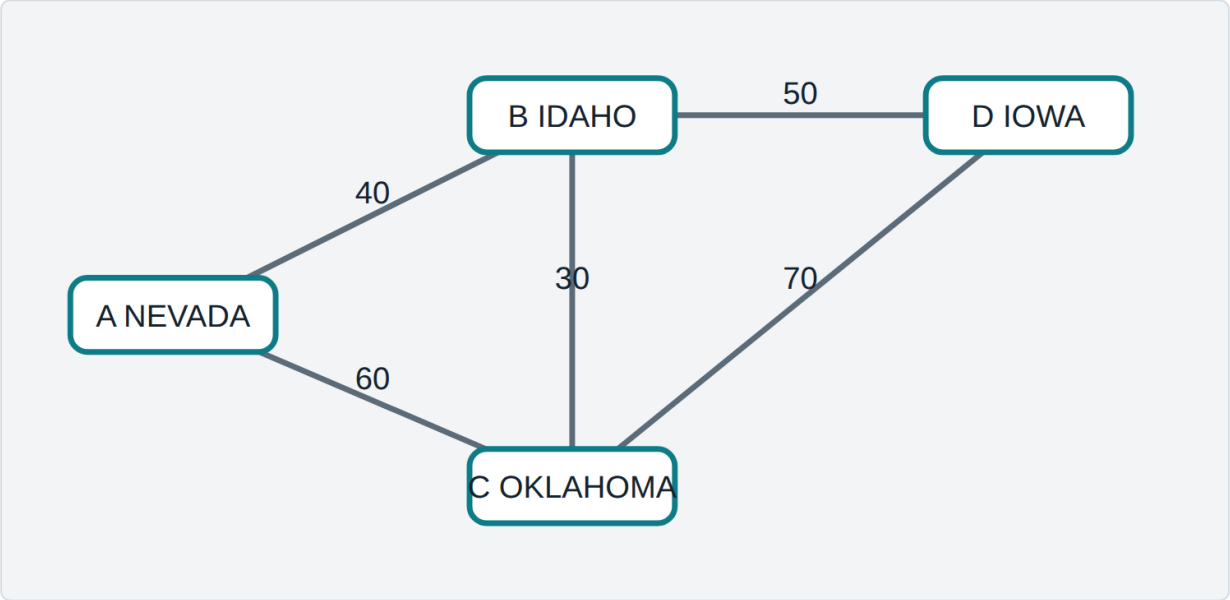
D. It uses a reference Bandwidth and the actual bandwidth of the connected link to calculate the route metric

Source answers: C

ccna-16-mi-q004 | EIGRP Questions

Provided HTML: CCNA 200-301 Questions.html | Question 4

Which two actions influence the EIGRP route selection process? (Choose two)



A. The router calculates the feasible distance of all paths to the destination route

B. The advertised distance is calculated by a downstream neighbor to inform the local router of the bandwidth on the link

C. The router calculates the reported distance by multiplying the delay on the exiting interface by 256

D. The router calculates the best backup path to the destination route and assigns it as the feasible successor

E. The router must use the advertised distance as the metric for any given route

Source answers: A, D

Source explanation

The reported distance (or advertised distance) is the cost from the neighbor to the destination. It is calculated from the router advertising the route to the network. For example in the topology below, suppose routerA & answer 'The router calculates the best backup path to the destination route and assigns it as the feasible successor' are exchanging their routing tables for the first time. Router answer 'The router calculates the best backup path to the destination route and assigns it as the feasible successor' says “Hey, the best metric (cost) from me to IOWA is 50 and the metric from you to IOWA is 90” and advertises it to routerA. Router answer 'The router calculates the reported distance by multiplying the delay on the exiting interface by 256' considers the first metric (50) as the Advertised distance. The second metric (90), which is from NEVADA to IOWA (through IDAHO), is called the Feasible distance.

The reported distance is calculated in the same way of calculating the metric. By default (K1 = 1, K2 = 0, K3 = 1, K4 = 0, K5 = 0), the metric is calculated as follows:

metric = [10,000,000 / slowest bandwidth [in kbps] + sum of delay [in µsec] / 10] * 256

-> Answer 'The router calculates the reported distance by multiplying the delay on the exiting interface by 256' is not correct.

Feasible successor is the backup route. To be a feasible successor, the route must have an Advertised distance (AD) less than the Feasible distance (FD) of the current successor route -> Answer 'The router calculates the best backup path to the destination route and assigns it as the feasible successor' is correct.

Feasible distance (FD): The sum of the AD plus the cost between the local router and the next-hop router. The router must calculate the FD of all paths to choose the best path to put into the routing table.

Note: Although the new CCNA exam does not have EIGRP topic but you should learn the basic knowledge of this routing protocol.

ccna-19-mi-q001 | Syslog Questions

Provided HTML: CCNA 200-301 Questions.html | Question 1

If a notice-level messaging is sent to a syslog server, which event has occurred?

A. An ARP Inspection has failed

B. A network device has restarted

C. A routing instance has flapped

D. A debug operation is running

Source answers: C

Source explanation

Usually no action is required when a route flaps so it generates the notification syslog level message (level 5).

ccna-19-mi-q002 | Syslog Questions

Provided HTML: CCNA 200-301 Questions.html | Question 2

What will happen if you configure the logging trap debug command on a router?

A. It causes the router to send all messages to the syslog server

B. It causes the router to stop sending all messages to the syslog server

C. It causes the router to send all messages with the severity levels Warning, Error, Critical, and Emergency to the syslog server

D. It causes the router to send messages with lower severity levels to the syslog server

Source answers: A

ccna-19-mi-q003 | Syslog Questions

Provided HTML: CCNA 200-301 Questions.html | Question 3

What event has occurred if a router sends a notice level message to a syslog server?

A. An ICMP connection has been built

B. An interface line has changed status

C. A certificate has expired

D. A TCP connection has been torn down

Source answers: B

Source explanation

If you used to configure a Cisco device then maybe you saw this notice level message:

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

Number 5 in the above message represents for the Notification Level (Normal, but significant, conditions exist).

ccna-22-mi-q001 | AAA Questions

Provided HTML: CCNA 200-301 Questions.html | Question 1

Which effect does the aaa new-model configuration command have?

A. It enables AAA services on the device.

B. It configures the device to connect to a RADIUS server for AAA.

C. It associates RADIUS server to an AAA group.

D. It configures a local user on the device.

Source answers: A

ccna-22-mi-q002 | AAA Questions

Provided HTML: CCNA 200-301 Questions.html | Question 2

How do AAA operations compare regarding user identification, user services and access control?

A. Authentication identifies users and accounting tracks user services

B. Authorization provides access control and authentication tracks user services

C. Authorization identifies users and authentication provides access control

D. Accounting tracks user services, and authentication provides access control

Source answers: A

ccna-22-mi-q003 | AAA Questions

Provided HTML: CCNA 200-301 Questions.html | Question 3

AAA stands for authentication, authorization, and accounting.

A. True

B. False

Source answers: A

ccna-22-mi-q004 | AAA Questions

Provided HTML: CCNA 200-301 Questions.html | Question 4

What is a difference between RADIUS and TACACS+?

A. RADIUS logs all commands that are entered by the administrator, but TACACS+ logs only start, stop, and interim commands

B. TACACS+ separates authentication and authorization, and RADIUS merges them

C. RADIUS is most appropriate for dial authentication, but TACACS+ can be used for multiple types of authentication

D. TACACS+ encrypts only password information and RADIUS encrypts the entire payload

Source answers: B

ccna-22-mi-q005 | AAA Questions

Provided HTML: CCNA 200-301 Questions.html | Question 5

Refer to the exhibit. Which password must an engineer use to enter the enable mode?

```
Atlanta#config t
Atlanta(config)#aaa new-model
Atlanta(config)#aaa authentication login default local
Atlanta(config)#line vty 0 4
Atlanta(config-line)#login authentication default
Atlanta(config-line)#exit
Atlanta(config)#username ciscoadmin password adminadmin123
Atlanta(config)#username ciscoadmin privilege 15
Atlanta(config)#enable password cisco123
Atlanta(config)#enable secret testing1234
Atlanta(config)#end
```

A. testing1234

B. cisco123

C. default

D. adminadmin123

Source answers: A

Source explanation

If neither the enable password command nor the enable secret command is configured, and if there is a line password configured for the console, the console line password serves as the enable password for all VTY sessions -> The “enable secret” will be used first if available, then “enable password” and line password.

ccna-22-mi-q006 | AAA Questions

Provided HTML: CCNA 200-301 Questions.html | Question 6

What is the primary difference between AAA authentication and authorization?

A. Authentication verifies a username and password, and authorization handles the communication between the authentication agent and the user database

B. Authentication controls the system processes a user can access and authorization logs the activities the user initiates

C. Authentication identifies and verifies a user who is attempting to access a system, and authorization controls the tasks the user can perform

D. Authentication identifies a user who is attempting to access a system, and authorization validates the users password

Source answers: C

Source explanation

AAA stands for Authentication, Authorization and Accounting.

+ Authentication: Specify who you are (usually via login username & password)
+ Authorization: Specify what actions you can do, what resource you can access
+ Accounting: Monitor what you do, how long you do it (can be used for billing and auditing)

An example of AAA is shown below:

+ Authentication: “I am a normal user. My username/password is user_tom/learnforever“
+ Authorization: “user_tom can access LearnCCNA server via HTTP and FTP“
+ Accounting: “user_tom accessed LearnCCNA server for 2 hours“. This user only uses “show” commands.

ccna-25-mi-q001 | DAI Questions

Provided HTML: CCNA 200-301 Questions.html | Question 1

Refer to the exhibit. What is the effect of this configuration?

```
ip arp inspection vlan 2
interface fastethernet 0/1
 switchport mode access
 switchport access vlan 2
```

A. All ingress and egress traffic is dropped because the interface is untrusted

B. Egress traffic is passed only if the destination is a DHCP server.

C. The switch discard all ingress ARP traffic with invalid MAC-to-IP address bindings

D. All ARP packets are dropped by the switch

Source answers: C

Source explanation

Dynamic ARP inspection is an ingress security feature; it does not perform any egress checking.

ccna-25-mi-q002 | DAI Questions

Provided HTML: CCNA 200-301 Questions.html | Question 2

Refer to the exhibit. What is the effect of this configuration?

```
ip arp inspection vlan 2
interface fastethernet 0/1
 switchport mode access
 switchport access vlan 2
```

A. Dynamic ARP inspection is disabled because the ARP ACL is missing

B. The switch port interface trust state becomes untrusted

C. The switch port remains administratively down until the interface is connected to another switch

D. The switch port remains down until it is configured to trust or untrust incoming packets

Source answers: B

Source explanation

Dynamic ARP inspection (DAI) is a security feature that validates ARP packets in a network. It intercepts, logs, and discards ARP packets with invalid IP-to-MAC address bindings. This capability protects the network from certain man-in-the-middle attacks. After enabling DAI, all ports become untrusted ports.

ccna-28-mi-q001 | QoS Questions

Provided HTML: CCNA 200-301 Questions.html | Question 1

Which two actions are performed by the Weighted Random Early Detection mechanism? (Choose two)

A. It supports protocol discovery

B. It guarantees the delivery of high-priority packets

C. It drops lower-priority packets before it drops higher-priority packets

D. It can identify different flows with a high level of granularity

E. It can mitigate congestion by preventing the queue from filling up

Source answers: C, E

Source explanation

Weighted Random Early Detection (WRED) is just a congestion avoidance mechanism. WRED drops packets selectively based on IP precedence. Edge routers assign IP precedences to packets as they enter the network. When a packet arrives, the following events occur:

1. The average queue size is calculated.
2. If the average is less than the minimum queue threshold, the arriving packet is queued.
3. If the average is between the minimum queue threshold for that type of traffic and the maximum threshold for the interface, the packet is either dropped or queued, depending on the packet drop probability for that type of traffic.
4. If the average queue size is greater than the maximum threshold, the packet is dropped.

WRED reduces the chances of tail drop (when the queue is full, the packet is dropped) by selectively dropping packets when the output interface begins to show signs of congestion (thus it can mitigate congestion by preventing the queue from filling up). By dropping some packets early rather than waiting until the queue is full, WRED avoids dropping large numbers of packets at once and minimizes the chances of global synchronization. Thus, WRED allows the transmission line to be used fully at all times.

WRED generally drops packets selectively based on IP precedence. Packets with a higher IP precedence are less likely to be dropped than packets with a lower precedence. Thus, the higher the priority of a packet, the higher the probability that the packet will be delivered (-> answer 'It drops lower-priority packets before it drops higher-priority packets' is correct).

Reference: https://www.cisco.com/c/en/us/td/docs/ios-xml/ios/qos_conavd/configuration/15-mt/qos-conavd-15-mt-book/qos-conavd-cfg-wred.html

ccna-28-mi-q002 | QoS Questions

Provided HTML: CCNA 200-301 Questions.html | Question 2

Which two QoS tools can provide congestion management? (Choose two)

A. PQ

B. CBWFQ

C. PBR

D. CAR

E. FRTS

Source answers: A, B

Source explanation

This module discusses the types of queueing and queueing-related features (such as bandwidth management) which constitute the congestion management QoS features:

Class-based WFQ (CBWFQ): extends the standard WFQ functionality to provide support for user-defined traffic classes. For CBWFQ, you define traffic classes based on match criteria including protocols, access control lists (ACLs), and input interfaces. Packets satisfying the match criteria for a class constitute the traffic for that class.

Priority queueing (PQ): With PQ, packets belonging to one priority class of traffic are sent before all lower priority traffic to ensure timely delivery of those packets.

Reference: https://www.cisco.com/c/en/us/td/docs/ios-xml/ios/qos_conmgt/configuration/xe-3s/qos-conmgt-xe-3s-book/qos-conmgt-oview.html

Note: Committed Access Rate (CAR) is only used for bandwidth limitation by dropping excessive traffic.

ccna-28-mi-q003 | QoS Questions

Provided HTML: CCNA 200-301 Questions.html | Question 3

Which feature can you implement to reserve bandwidth for VoIP calls across the call path?

A. PQ

B. CBWFQ

C. RSVP

D. round robin

Source answers: C

Source explanation

The Resource Reservation Protocol (RSVP) protocol allows applications to reserve bandwidth for their data flows. It is used by a host, on the behalf of an application data flow, to request a specific amount of bandwidth from the network. RSVP is also used by the routers to forward bandwidth reservation requests.

ccna-28-mi-q004 | QoS Questions

Provided HTML: CCNA 200-301 Questions.html | Question 4

What does traffic shaping do to reduce congestion in a network?

A. drops packets

B. queues without buffering packets

C. buffers without queuing packets

D. buffers and queues packets

Source answers: D

Source explanation

The following diagram illustrates the key difference between traffic policing and traffic shaping. Traffic policing propagates bursts. When the traffic rate reaches the configured maximum rate (or committed information rate), excess traffic is dropped (or remarked). The result is an output rate that appears as a saw-tooth with crests and troughs. In contrast to policing, traffic shaping retains excess packets in a queue and then schedules the excess for later transmission over increments of time. The result of traffic shaping is a smoothed packet output rate.
Note: Committed information rate (CIR): The minimum guaranteed data transfer rate agreed to by the routing device.

ccna-28-mi-q005 | QoS Questions

Provided HTML: CCNA 200-301 Questions.html | Question 5

Which option is the main function of congestion management?

A. providing long-term storage of buffered data

B. discarding excess traffic

C. classifying traffic

D. queuing traffic based on priority

Source answers: D

ccna-28-mi-q006 | QoS Questions

Provided HTML: CCNA 200-301 Questions.html | Question 6

Which feature or protocol determines whether the QoS on the network is sufficient to support IP services?

A. LLDP

B. EEM

C. IPSLA

D. CDP

Source answers: C

Source explanation

IP SLA allows an IT professional to collect information about network performance in real time. Therefore it helps determine whether the QoS on the network is sufficient for IP services or not.

Cisco IOS Embedded Event Manager (EEM) is a powerful and flexible subsystem that provides real-time network event detection and onboard automation. It gives you the ability to adapt the behavior of your network devices to align with your business needs.

ccna-28-mi-q007 | QoS Questions

Provided HTML: CCNA 200-301 Questions.html | Question 7

What is the purpose of traffic shaping?

A. to provide fair queuing for buffered flows

B. be a marking mechanism that identifies different flows

C. to limit the bandwidth that a flow can use

D. to mitigate delays over slow links

Source answers: C

Source explanation

The primary reasons you would use traffic shaping are to control access to available bandwidth, to ensure that traffic conforms to the policies established for it, and to regulate the flow of traffic in order to avoid congestion that can occur when the sent traffic exceeds the access speed of its remote, target interface.

ccna-31-mi-q001 | Wireless Questions 2

Provided HTML: CCNA 200-301 Questions.html | Question 1

Which WPA3 enhancement protects against hackers viewing traffic on the Wi-Fi network?

A. SAE encryption

B. TKIP encryption

C. AES encryption

D. Scrambled encryption key

Source answers: A

Source explanation

Wi-Fi Protected Access version 3 (WPA3) is a new Wi-Fi Alliance’s (WFA) security standard for personal and enterprise networks. It aims to improve overall Wi-Fi security by using modern security algorithms and stronger cipher suites. WPA3 has two parts:

+ WPA3-Personal: Uses simultaneous authentication of equals (SAE) instead of pre-shared key (PSK), providing users with stronger security protections against attacks such as offline dictionary attacks, key recovery, and message forging.
+ WPA3-Enterprise: Offers stronger authentication and link-layer encryption methods, and an optional 192-bit security mode for sensitive security environments.

ccna-31-mi-q002 | Wireless Questions 2

Provided HTML: CCNA 200-301 Questions.html | Question 2

What is a recommended approach to avoid co-channel congestion while installing access points that use the 2.4 GHz frequency?

A. one nonoverlapping channel

B. one overlapping channel

C. different nonoverlapping channels

D. different overlapping channels

Source answers: C

Source explanation

In the 2.4 GHz band, 1, 6, and 11 are the only non-overlapping channels so they should be chosen while installing APs.

ccna-31-mi-q003 | Wireless Questions 2

Provided HTML: CCNA 200-301 Questions.html | Question 3

Using direct sequence spread spectrum, which three 2.4-GHz channels are used to limit collisions?

A. 1,6,11

B. 1,2,3

C. 1,5,10

D. 5,6,7

Source answers: A

Source explanation

In the 2.4 GHz band, 1, 6, and 11 are the only non-overlapping channels.

ccna-31-mi-q004 | Wireless Questions 2

Provided HTML: CCNA 200-301 Questions.html | Question 4

Which protocol prompts the Wireless LAN Controller to generate its own local web administration SSL certificate for GUI access?

A. TACACS+

B. HTTP

C. HTTPS

D. RADIUS

Source answers: C

Source explanation

When you enable HTTPS, the controller generates its own local web administration SSL certificate and automatically applies it to the GUI.

Reference: https://www.cisco.com/c/en/us/td/docs/wireless/controller/8-0/configuration-guide/b_cg80/b_cg80_chapter_011.html

ccna-31-mi-q005 | Wireless Questions 2

Provided HTML: CCNA 200-301 Questions.html | Question 5

What is a difference between local AP mode and FlexConnect AP mode?

A. Local AP mode causes the AP to behave as if it were an autonomous AP

B. FlexConnect AP mode fails to function if the AP loses connectivity with the WLC

C. FlexConnect AP mode bridges the traffic from the AP to the WLC when local switching is configured

D. Local AP mode creates two CAPWAP tunnels per AP to the WLC

Source answers: D

Source explanation

In Central Switched (Local) mode, an AP creates two CAPWAP tunnels to the Wireless Controller. One tunnel is used for forwarding data traffic and the other is used for forwarding the management traffic.

ccna-31-mi-q006 | Wireless Questions 2

Provided HTML: CCNA 200-301 Questions.html | Question 6

What is a function of Wireless LAN Controller?

A. use SSIDs to distinguish between wireless clients

B. register with a single access point that controls traffic between wired and wireless endpoints

C. monitor activity on wireless and wired LANs

D. send LWAPP packets to access points

Source answers: D

ccna-31-mi-q007 | Wireless Questions 2

Provided HTML: CCNA 200-301 Questions.html | Question 7

A wireless administrator has configured a WLAN; however, the clients need access to a less congested 5-GHz network for their voice quality. What action must be taken to meet the requirement?

A. enable DTIM

B. enable AAA override

C. enable Band Select

D. enable RX-SOP

Source answers: C

Source explanation

Band select or Band Direction is a new feature that encourages dual band clients to connect to 5 G-Hz networks. Band select is disabled by default. The Band Select function provides a better wireless experience for users.

ccna-31-mi-q008 | Wireless Questions 2

Provided HTML: CCNA 200-301 Questions.html | Question 8

How does CAPWAP communicate between an access point in local mode and a WLC?

A. The access point must directly connect to the WLC using a copper cable

B. The access point must be connected to the same switch as the WLC

C. The access point has the ability to link to any switch in the network, assuming connectivity to the WLC

D. The access point must not be connected to the wired network, as it would create a loop

Source answers: C

ccna-31-mi-q009 | Wireless Questions 2

Provided HTML: CCNA 200-301 Questions.html | Question 9

When a WPA2-PSK WLAN is configured in the Wireless LAN Controller, what is the minimum number of characters that is required for the passphrase?

A. 18

B. 6

C. 8

D. 12

Source answers: C

Source explanation

WPA/WPA2 preshared keys must contain 8 to 63 ASCII text characters or 64 hexadecimal characters.

ccna-34-mi-q001 | Drag Drop Questions

Provided HTML: CCNA 200-301 Questions.html | Question 1

Refer to the exhibit.

```
[root#HostTime=]#ip route
default via 192.168.1.193 dev eth1 proto static
192.168.1.0/26 dev sth1 proto kernel scope link src 192.168.1.200 metric 1

[root#HostTime=]#ip addr show eth1
eth1:mtu 1500 qdisc pfifo_fast qlan 1000
 link/ether 00:0C:22:83:79:A3 brd ff:ff:ff:ff:ff:ff
 inet 192.168.1.200/26 hrd 192.168.1.255 scope global eth1
 inet6 fe80::20c::29ff:fe89:79b3/64 scope link
 valid_lft forever preferred_lft forever
```

Drag and drop the networking parameters from the left onto the correct values on the right.

Items

00:0C:22

00:0C:22:83:79:A3

192.168.1.193

192.168.1.200

255.255.255.192

Categories

NIC vendor OUI

NIC MAC address

default gateway

host IP address

subnet mask

Source solution: 00:0C:22 -> NIC vendor OUI; 00:0C:22:83:79:A3 -> NIC MAC address; 192.168.1.193 -> default gateway; 192.168.1.200 -> host IP address; 255.255.255.192 -> subnet mask

Source explanation

+ 00:0C:22: NIC vendor OUI
+ 00:0C:22:83:79:A3: NIC MAC address
+ 192.168.1.193: default gateway
+ 192.168.1.200: host IP address
+ 255.255.255.192: subnet mask

The “ip route” and “ip addr show eth1” are Linux commands.
+ “ip route”: display the routing table
+ “ip addr show eth1”: get depth information (only on eth1 interface) about your network interfaces like IP Address, MAC Address information

ccna-34-mi-q002 | Drag Drop Questions

Provided HTML: CCNA 200-301 Questions.html | Question 2

Drag and drop the functions from the left onto the correct network components on the right.

Items

1. holds the TCP/IP settings to be distributed to the clients

2. resolves web URLs to IP addresses

3. stores a list of IP addresses mapped to names

4. assigns a default gateway to a client

5. assigns IP addresses to enabled clients

Categories

DHCP Server

DNS Server

Source solution: 1. holds the TCP/IP settings to be distributed to the clients -> DHCP Server; 2. resolves web URLs to IP addresses -> DNS Server; 3. stores a list of IP addresses mapped to names -> DNS Server; 4. assigns a default gateway to a client -> DHCP Server; 5. assigns IP addresses to enabled clients -> DHCP Server

ccna-34-mi-q003 | Drag Drop Questions

Provided HTML: CCNA 200-301 Questions.html | Question 3

Drag the OSPF parameters to the correct places on the right. Note: There is one unused answer in this question.

Items

1. netmask

2. OSPF process ID

3. router ID

4. IP address

5. area ID

6. timers

Categories

must be unique

must match

unused answer

Source solution: 1. netmask -> must match; 2. OSPF process ID -> unused answer; 3. router ID -> must be unique; 4. IP address -> must be unique; 5. area ID -> must match; 6. timers -> must match

ccna-34-mi-q004 | Drag Drop Questions

Provided HTML: CCNA 200-301 Questions.html | Question 4

Drag and drop the IPv4 network subnets from the left onto the correct usable host ranges on the right.

Items

172.28.228.144/18

172.28.228.144/21

172.28.228.144/23

172.28.228.144/25

172.28.228.144/29

Categories

172.28.192.1 – 172.28.255.254

172.28.224.1 – 172.28.231.254

172.28.228.1 – 172.28.229.254

172.28.228.129 – 172.28.228.254

172.28.228.145 – 172.28.228.150

Source solution: 172.28.228.144/18 -> 172.28.192.1 – 172.28.255.254; 172.28.228.144/21 -> 172.28.224.1 – 172.28.231.254; 172.28.228.144/23 -> 172.28.228.1 – 172.28.229.254; 172.28.228.144/25 -> 172.28.228.129 – 172.28.228.254; 172.28.228.144/29 -> 172.28.228.145 – 172.28.228.150

Source explanation

+ 172.28.228.144/18: 172.28.192.1 – 172.28.255.254
+ 172.28.228.144/21: 172.28.224.1 – 172.28.231.254
+ 172.28.228.144/23: 172.28.228.1 – 172.28.229.254
+ 172.28.228.144/25: 172.28.228.129 – 172.28.228.254
+ 172.28.228.144/29: 172.28.228.145 – 172.28.228.150

This subnet question requires us to grasp how to subnet very well. To quickly find out the subnet range, we have to find out the increment and the network address of each subnet. Let’s take an example with the subnet 172.28.228.144/18:

From the /18 (= 1100 0000 in the 3rd octet), we find out the increment is 64. Therefore the network address of this subnet must be the greatest multiple of the increment but not greater than the value in the 3rd octet (228). We can find out the 3rd octet of the network address is 192 (because 192 = 64 * 3 and 192 < 228) -> The network address is 172.28.192.0. So the first usable host should be 172.28.192.1 and it matches with the 5th answer on the right. In this case we don’t need to calculate the broadcast address because we found the correct answer.

Let’s take another example with subnet 172.28.228.144/23 -> The increment is 2 (as /23 = 1111 1110 in 3rd octet). The 3rd octet of the network address is 228 (because 228 is the multiply of 2 and equal to the 3rd octet) -> The network address is 172.28.228.0 -> The first usable host is 172.28.228.1. It is not necessary but if we want to find out the broadcast address of this subnet, we can find out the next network address, which is 172.28.(228 + the increment number).0 or 172.28.230.0 then reduce 1 bit -> 172.28.229.255 is the broadcast address of our subnet. Therefore the last usable host is 172.28.229.254.

If you are still unclear about how to do subnetting quickly, please read part 3 of our “Subnetting Tutorial – Subnetting Made Easy” at https://www.9tut.com/subnetting-tutorial/3

ccna-34-mi-q005 | Drag Drop Questions

Provided HTML: CCNA 200-301 Questions.html | Question 5

Drag and drop the threat-mitigation techniques from the left onto the types of threat or attack they mitigate on the right.

Items

802.1q double tagging

ARP spoofing

unwanted superior BPDUs

unwanted BPDUs on PortFast-enabled interfaces

Categories

Configure a VLAN access control list

Configure the dynamic ARP inspection feature

Configure the root guard feature

Configure the BPDU guard feature

Source solution: 802.1q double tagging -> Configure a VLAN access control list; ARP spoofing -> Configure the dynamic ARP inspection feature; unwanted superior BPDUs -> Configure the root guard feature; unwanted BPDUs on PortFast-enabled interfaces -> Configure the BPDU guard feature

Source explanation

+ 802.1q double tagging: Configure VACL
+ ARP spoofing: Configure dynamic ARP inspection
+ unwanted superior BPDUs: Configure root guard
+ unwanted BPDUs on PortFast-enabled interfaces: Configure BPDU guard

Double-Tagging attack:

Attacker --VLAN 10--> Company Switch A --Native VLAN 10--> Company Switch B --VLAN 20--> Victim
[Payload|VLAN 20|VLAN 10] [Payload|VLAN 20] [Payload]

In this attack, the attacking computer generates frames with two 802.1Q tags. The first tag matches the native VLAN of the trunk port (VLAN 10 in this case), and the second matches the VLAN of a host it wants to attack (VLAN 20).

When the packet from the attacker reaches Switch A, Switch A only sees the first VLAN 10 and it matches with its native VLAN 10 so this VLAN tag is removed. Switch A forwards the frame out all links with the same native VLAN 10. Switch B receives the frame with an tag of VLAN 20 so it removes this tag and forwards out to the Victim computer.

Note: This attack only works if the trunk (between two switches) has the same native VLAN as the attacker.

To mitigate this type of attack, you can use VLAN access control lists (VACLs, which applies to all traffic within a VLAN. We can use VACL to drop attacker traffic to specific victims/servers) or implement Private VLANs.

ARP attack (like ARP poisoning/spoofing) is a type of attack in which a malicious actor sends falsified ARP messages over a local area network as ARP allows a gratuitous reply from a host even if an ARP request was not received. This results in the linking of an attacker’s MAC address with the IP address of a legitimate computer or server on the network. This is an attack based on ARP which is at Layer 2. Dynamic ARP

inspection (DAI) is a security feature that validates ARP packets in a network which can be used to mitigate this type of attack.

ccna-34-mi-q006 | Drag Drop Questions

Provided HTML: CCNA 200-301 Questions.html | Question 6

Drag and drop the description of file-transfer protocols from the left onto the correct protocols on the right.

Items

1. does not require user authentication

2. uses UDP

3. uses TCP

4. uses ports 20 and 21

5. provides reliability when loading an IOS image upon boot up

6. uses port 69

Categories

FTP

TFTP

Source solution: 1. does not require user authentication -> TFTP; 2. uses UDP -> TFTP; 3. uses TCP -> FTP; 4. uses ports 20 and 21 -> FTP; 5. provides reliability when loading an IOS image upon boot up -> FTP; 6. uses port 69 -> TFTP

ccna-34-mi-q007 | Drag Drop Questions

Provided HTML: CCNA 200-301 Questions.html | Question 7

Drag and drop the application protocols from the left onto the suitable transport protocols on the right.

Items

1. SMTP

2. SNMP

3. DHCP

4. TFTP

5. FTP

6. SSH

Categories

TCP

UDP

Source solution: 1. SMTP -> TCP; 2. SNMP -> UDP; 3. DHCP -> UDP; 4. TFTP -> UDP; 5. FTP -> TCP; 6. SSH -> TCP

ccna-34-mi-q008 | Drag Drop Questions

Provided HTML: CCNA 200-301 Questions.html | Question 8

Drag and drop the descriptions of IP protocol transmissions from the left onto the correct IP traffic types on the right.

Items

1. Sends Transmission in Sequence

2. Transmits packets individually

3. Uses higher transmission rate to support latency-sensitive apps

4. Uses a lower transmission rate to ensure reliability

5. Transmissions include an 8-byte header

6. Transmits packet as stream

Categories

TCP

UDP

Source solution: 1. Sends Transmission in Sequence -> TCP; 2. Transmits packets individually -> UDP; 3. Uses higher transmission rate to support latency-sensitive apps -> UDP; 4. Uses a lower transmission rate to ensure reliability -> TCP; 5. Transmissions include an 8-byte header -> UDP; 6. Transmits packet as stream -> TCP

Source explanation

UDP is a simpler message-based connectionless protocol. In connectionless protocols, there is no effort made to setup a dedicated end-to-end connection.

...

Datagrams - Packets are sent individually and are guaranteed to be whole if they arrive. Packets have definite bounds and no split or merge into data streams may exist.

According to https://community.cisco.com/t5/networking-documents/udp/ta-p/3114870

TCP is connection-oriented and UDP is connectionless. This means that before sending TCP packets, a connection is established between the server and the client. This process of setting up a connection is called TCP handshaking. The stream of packets is then sent over this connection -> TCP transmits packet as stream.

Reference: https://www.vpnmentor.com/blog/tcp-vs-udp/

ccna-34-mi-q009 | Drag Drop Questions

Provided HTML: CCNA 200-301 Questions.html | Question 9

Drag and drop the WLAN components from the left onto the correct descriptions on the right.

Items

device that manages access points

device that provides Wi-Fi devices with a connection to a wired network

used for out of band management of a WLC

used to support mobility management of the WLC

applied to the WLAN for wireless client communication

Categories

wireless LAN controller

access point

service port

virtual interface

dynamic interface

Source solution: device that manages access points -> wireless LAN controller; device that provides Wi-Fi devices with a connection to a wired network -> access point; used for out of band management of a WLC -> service port; used to support mobility management of the WLC -> virtual interface; applied to the WLAN for wireless client communication -> dynamic interface

Source explanation

+ device that manages access points: wireless LAN controller
+ device that provides Wi-Fi devices with a connection to a wired network: access point
+ used for out of band management of a WLC: service port
+ used to support mobility management of the WLC: virtual interface
+ applied to the WLAN for wireless client communication: dynamic interface

The service port can be used management purposes, primarily for out-of-band management. However, AP management traffic is not possible across the service port. In most cases, the service port is used as a “last resort” means of accessing the controller GUI for management purposes. For example, in the case where the system distribution ports on the controller are down or their communication to the wired network is otherwise degraded.

A dynamic interface with the Dynamic AP Management option enabled is used as the tunnel source for packets from the controller to the access point and as the destination for CAPWAP packets from the access point to the controller.

The virtual interface is used to support mobility management, Dynamic Host Configuration Protocol (DHCP) relay, and embedded Layer 3 security such as guest web authentication. It also maintains the DNS gateway host name used by Layer 3 security and mobility managers to verify the source of certificates when Layer 3 web authorization is enabled.

Reference:
https://www.cisco.com/c/en/us/td/docs/wireless/controller/8-5/config-guide/b_cg85/ports_and_interfaces.html

ccna-34-mi-q010 | Drag Drop Questions

Provided HTML: CCNA 200-301 Questions.html | Question 10

Drag and drop the networking parameters from the left on to the correct values on the right.

Items

1. SMTP

2. SNMP

3. TFTP

4. VoIP

5. SSH

6. FTP

Categories

Connection Oriented

Connectionless

Source solution: 1. SMTP -> Connection Oriented; 2. SNMP -> Connectionless; 3. TFTP -> Connectionless; 4. VoIP -> Connectionless; 5. SSH -> Connection Oriented; 6. FTP -> Connection Oriented

Source explanation

SSH uses TCP port 22 while SNMP uses UDP port 161 and 162.

ccna-34-mi-q011 | Drag Drop Questions

Provided HTML: CCNA 200-301 Questions.html | Question 11

Drag drop the descriptions from the left on to the correct configuration-management technologies on the right.

Items

1. uses Ruby for fundamental configuration elements

2. uses TCP port 10002 for configuration push jobs

3. uses SSH for remote device communication

4. fundamental configuration elements are stored in a manifest

5. uses TCP 8140 for communication

6. uses YAML for fundamental configuration elements

Categories

Ansible

Chef

Puppet

Source solution: 1. uses Ruby for fundamental configuration elements -> Chef; 2. uses TCP port 10002 for configuration push jobs -> Chef; 3. uses SSH for remote device communication -> Ansible; 4. fundamental configuration elements are stored in a manifest -> Puppet; 5. uses TCP 8140 for communication -> Puppet; 6. uses YAML for fundamental configuration elements -> Ansible

Source explanation

The focus of Ansible is to be streamlined and fast, and to require no node agent installation. Thus, Ansible performs all functions over SSH. Ansible is built on Python, in contrast to the Ruby foundation of Puppet and Chef.

TCP port 10002 is the command port. It may be configured in the Chef Push Jobs configuration file . This port allows Chef Push Jobs clients to communicate with the Chef Push Jobs server.

Puppet is an open-source configuration management solution, which is built with Ruby and offers custom Domain Specific Language (DSL) and Embedded Ruby (ERB) templates to create custom Puppet language files, offering a declarative-paradigm programming approach.

A Puppet piece of code is called a manifest, and is a file with .pp extension.

ccna-34-mi-q012 | Drag Drop Questions

Provided HTML: CCNA 200-301 Questions.html | Question 12

Drag and drop the AAA functions from the left onto the correct AAA services on the right.

Items

1. records user activities

2. restricts the services that are available to a user

3. identifies the user

4. controls the actions that a user can perform

5. provides analytical information for the network administrator

6. verifies the password associated with a user

Categories

Authentication

Authorization

Accounting

Source solution: 1. records user activities -> Accounting; 2. restricts the services that are available to a user -> Authorization; 3. identifies the user -> Authentication; 4. controls the actions that a user can perform -> Authorization; 5. provides analytical information for the network administrator -> Accounting; 6. verifies the password associated with a user -> Authentication

ccna-34-mi-q013 | Drag Drop Questions

Provided HTML: CCNA 200-301 Questions.html | Question 13

Drag and Drop the benefits of a Cisco Wireless LAN Controller from the left onto the correct examples on the right.

Items

Dynamic RF Feature

Easy Deployment Process

Optimized user performance

Easy upgrade process

Categories

Access points auto adjust signal strength

Controller provides centralized management of users and VLANs

Controller uses loadbalancing to maximize throughput

Controller image auto deployed to access Points

Source solution: Dynamic RF Feature -> Access points auto adjust signal strength; Easy Deployment Process -> Controller provides centralized management of users and VLANs; Optimized user performance -> Controller uses loadbalancing to maximize throughput; Easy upgrade process -> Controller image auto deployed to access Points

Source explanation

+ Dynamic RF Feature: Access points auto adjust signal strength
+ Easy Deployment Process: Controller provides centralized management of users and VLANs
+ Optimized user performance: Controller uses loadbalancing to maximize throughput
+ Easy upgrade process: Controller image auto deployed to access Points

ccna-34-mi-q014 | Drag Drop Questions

Provided HTML: CCNA 200-301 Questions.html | Question 14

Drag and drop the Cisco Wireless LAN Controller security settings from the left onto the correct security mechanism categories on the right.

Items

1. web policy

2. Passthrough

3. WPA+WPA2

4. 802.1X

Categories

Layer 2 Security Mechanism

Layer 3 Security Mechanisms (for WLAN)

Source solution: 1. web policy -> Layer 3 Security Mechanisms (for WLAN); 2. Passthrough -> Layer 3 Security Mechanisms (for WLAN); 3. WPA+WPA2 -> Layer 2 Security Mechanism; 4. 802.1X -> Layer 2 Security Mechanism

Source explanation

Layer 2 Security Mechanism includes WPA+WPA2, 802.1X, Static WEP, CKIP while Layer 3 Security Mechanisms (for WLAN) includes IPSec, VPN Pass-Through, Web Passthrough …

Reference: https://www.cisco.com/c/en/us/support/docs/wireless/4400-series-wireless-lan-controllers/106082-wlc-compatibility-matrix.html

ccna-37-mi-q001 | VPN Questions

Provided HTML: CCNA 200-301 Questions.html | Question 1

What is a function of a remote access VPN?

A. used exclusively when a user is connected to a company’s internal network

B. allows the users to access company internal network resources through a secure tunnel

C. establishes a secure tunnel between two branch sites

D. used cryptographic tunneling to protect the privacy of data for multiple users simultaneously

Source answers: B

ccna-37-mi-q002 | VPN Questions

Provided HTML: CCNA 200-301 Questions.html | Question 2

Which type of VPN uses a hub-and-spoke configuration to establish a full mesh topology?

A. dynamic multipoint VPN

B. MPLS VPN

C. GRE over IPsec

D. IPsec virtual tunnel interface

Source answers: A

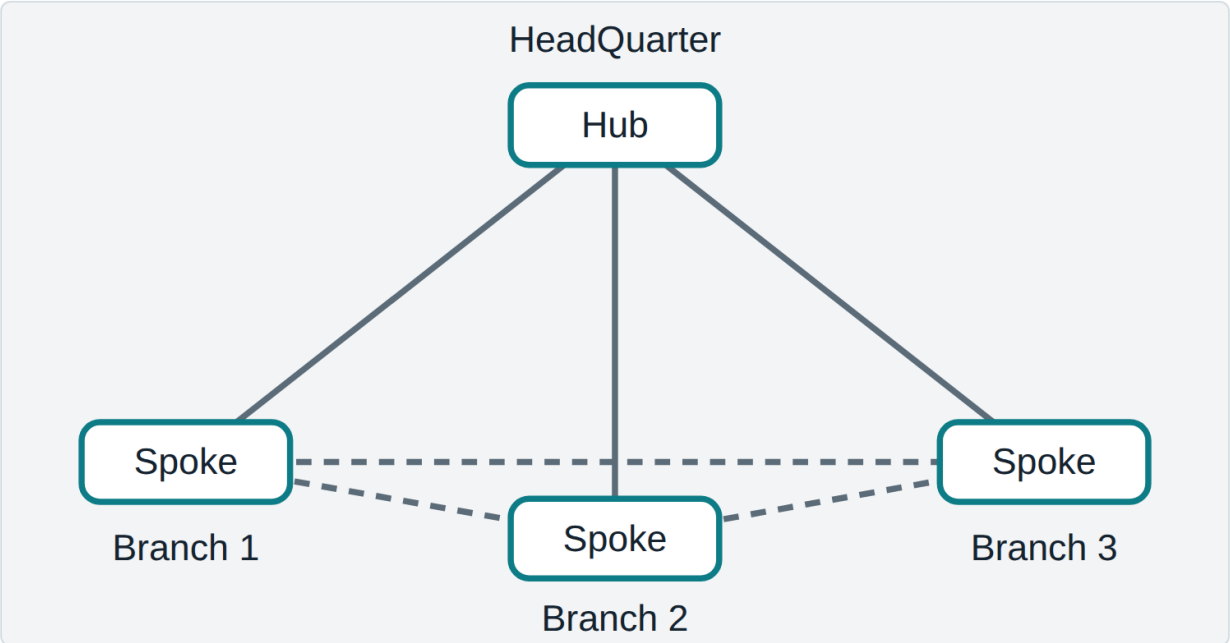
Source explanation

Dynamic Multipoint VPN (DMVPN) is a solution of Cisco. DMVPN provides the following advantages:

+ Provides full meshed connectivity with simple Hub-and-Spoke topology. The spokes can communicate between each other without going through Hub
+ Only one static public IP address is required on Hub. Spokes can use dynamic (unknown) public IP addresses.
+ The configuration is simple even in large network. No additional configuration is required on Hub when new Spokes are added.
HubSpokeSpokeSpokeHeadQuarterBranch 1Branch 2Branch 3

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ccna-37-mi-q003 | VPN Questions

Provided HTML: CCNA 200-301 Questions.html | Question 3

What mechanism carries multicast traffic between remote sites and supports encryption?

A. IPsec over ISATAP

B. ISATAP

C. GRE

D. GRE over IPsec

Source answers: D

ccna-40-mi-q001 | Miscellaneous Questions

Provided HTML: CCNA 200-301 Questions.html | Question 1

Which technology must be implemented to configure network device monitoring with the highest security?

A. SNMPv3

B. NetFlow

C. syslog

D. IP SLA

Source answers: A

Source explanation

SNMPv3—The most up-to-date protocol focuses on security. SNMPv3 defines a security model, user-based security model (USM), and a view-based access control model (VACM). SNMPv3 USM provides data integrity, data origin authentication, message replay protection, and protection against disclosure of the message payload.

Reference: https://www.juniper.net/documentation/us/en/software/junos/network-mgmt/topics/topic-map/network-monitoring-by-using-snmp.html

ccna-40-mi-q002 | Miscellaneous Questions

Provided HTML: CCNA 200-301 Questions.html | Question 2

Which command can you enter to allow Telnet to be supported in addition to SSH?

A. transport input telnet

B. transport input telnet ssh

C. privilege level 15

D. no transport input telnet

Source answers: B

ccna-40-mi-q003 | Miscellaneous Questions

Provided HTML: CCNA 200-301 Questions.html | Question 3

What are two differences between optical-fiber cabling and copper cabling? (Choose two)

A. The glass core component is encased in a cladding

B. The data can pass through the cladding

C. Fiber connects to physical interfaces using RJ-45 connections

D. A BNC connector is used for fiber connections

E. Light is transmitted through the core of the fiber

Source answers: A, E

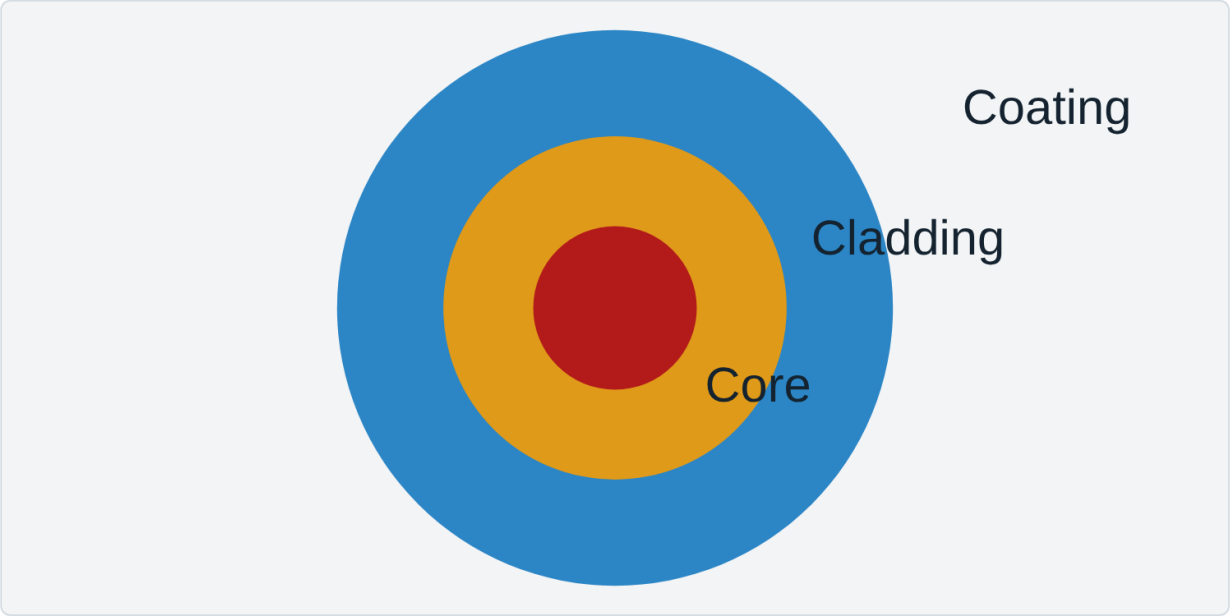
Source explanation

The two main elements of an optical fiber are its core and cladding. The “core”, or the axial part of the optical fiber made of silica glass, is the light transmission area of the fiber. It may sometimes be treated with a “doping” element to change its refractive index and therefore the velocity of light down the fiber.

The “cladding” is the layer completely surrounding the core.
CoatingCladdingCore
Reference: https://www.cisco.com/c/en/us/products/collateral/interfaces-modules/transceiver-modules/white_paper_c11-463661.html

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Reference: https://www.cisco.com/c/en/us/products/collateral/interfaces-modules/transceiver-modules/white_paper_c11-463661.html

ccna-40-mi-q004 | Miscellaneous Questions

Provided HTML: CCNA 200-301 Questions.html | Question 4

A device detects two stations transmitting frames at the same time. This condition occurs after the first 64 bytes of the frame is received interface counter increments?

A. CRC

B. runt

C. late collision

D. collision

Source answers: C

Source explanation

A late collision is defined as any collision that occurs after the first 512 bits (or 64th byte) of the frame have been transmitted. The usual possible causes are full-duplex/half-duplex mismatch, exceeded Ethernet cable length limits, or defective hardware such as incorrect cabling, non-compliant number of hubs in the network, or a bad NIC.

Late collisions should never occur in a properly designed Ethernet network. They usually occur when Ethernet cables are too long or when there are too many repeaters in the network.

Reference: https://www.cisco.com/en/US/docs/internetworking/troubleshooting/guide/tr1904.html

ccna-40-mi-q005 | Miscellaneous Questions

Provided HTML: CCNA 200-301 Questions.html | Question 5

A network engineer must back up 20 network router configurations globally within a customer environment. Which protocol allows the engineer to perform this function using the copy function?

A. SNMP

B. SMTP

C. COP

D. ARP

Source answers: A

Source explanation

SNMP is an application-layer protocol that provides a message format for communication between SNMP managers and agents. SNMP provides a standardized framework and a common language used for the monitoring and management of devices in a network.
The SNMP framework has three parts:
+ An SNMP manager
+ An SNMP agent
+ A Management Information Base (MIB)

The Management Information Base (MIB) is a virtual information storage area for network management information, which consists of collections of managed objects.

With SNMP, the network administrator can send commands to multiple routers to do the backup.

ccna-40-mi-q006 | Miscellaneous Questions

Provided HTML: CCNA 200-301 Questions.html | Question 6

A network analyst is tasked with configuring the date and time on a router using EXEC mode. The date must be set to January 1,2020 and the time must be set to 12:00 am. Which command should be used?

A. clock timezone

B. clock set

C. clock summer-time date

D. clock summer-time recurring

Source answers: B

Source explanation

In this example, the clock time is set to 12:00 am with the clock date of January 1,2020.

R1#clock set 12:00:00 jan 1 2020

ccna-40-mi-q007 | Miscellaneous Questions

Provided HTML: CCNA 200-301 Questions.html | Question 7

A Cisco IP phone receive untagged data traffic from an attached PC. Which action is taken by the phone?

A. It allows the traffic to pass through unchanged

B. It tags the traffic with the default VLAN

C. It drops the traffic

D. It tags the traffic with the native VLAN

Source answers: A

Source explanation

Untagged traffic from the device attached to the Cisco IP Phone passes through the phone unchanged, regardless of the trust state of the access port on the phone.

Reference: https://www.cisco.com/c/en/us/td/docs/switches/lan/catalyst2960/software/release/12-2_40_se/configuration/guide/scg/swvoip.pdf

ccna-40-mi-q008 | Miscellaneous Questions

Provided HTML: CCNA 200-301 Questions.html | Question 8

Which function does an SNMP agent perform?

A. It coordinates user authentication between a network device and a TACACS+ or RADIUS server

B. It requests information from remote network nodes about catastrophic system events.

C. It manages routing between Layer 3 devices in a network

D. It sends information about MIB variables in response to requests from the NMS

Source answers: D